

Optimal Linear Functional Estimation for Proportionally High-Dimensional Linear Models

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Abstract

We study estimation of a linear functional in random-design linear models when $\gamma = p/n \in (1, \infty)$ under the dense regime where the design covariance is unknown, and neither the regression vector nor the precision matrix is assumed sparse. Suppose the spectrum of the design covariance belongs to $[\ell, L]$ and $\kappa = L/\ell$, we establish the minimax mean squared rate for the linear functional has the order $n^{-\alpha(\kappa, \gamma)}$, where the exponent $\alpha(\kappa, \gamma)$ has an explicit elliptic-integral representation and $\alpha(\kappa, \gamma) < 1$ when both $\kappa > 1$ and $\gamma > 1$. We propose the Bocca estimator, which combines empirically calibrated resolvents through a complex contour integral to attain the optimal rate.

1 Introduction

Many regression problems concern a single coefficient or a prespecified contrast rather than the entire regression vector. Examples include an exposure coefficient after adjustment for many covariates and a contrast among effects in a genomic study. Such targets can remain scientifically meaningful when there is no convincing reason to assume that only a few covariates affect the response. This distinction motivates inference with dense regression coefficients (Zhu and Bradic, 2018a; Bradic et al., 2022; Sankaranarayanan et al., 2025). We study the linear model $Y = X\beta + \varepsilon$, with unknown design covariance $\Sigma = \text{Cov}(X)$ and unknown noise variance $\sigma^2 = \text{Var}(\varepsilon)$. The parameter of interest is the scalar functional $\beta_v = \langle v, \beta \rangle$, where v is a known deterministic unit vector. Most of the time, we consider the canonical vector $v = e_j$ for some $j \in \{1, \dots, p\}$ to estimate a single coefficient β_j . Our focus is the dense proportional regime in which $\gamma = p/n \in (1, \infty)$ and both the coefficient vector β and the precision matrix Σ^{-1} may be dense.

In this regime, recovering a low-dimensional functional and recovering the full vector are fundamentally different problems. Even under isotropic Gaussian design, the minimax ℓ_2 -norm rate over a fixed-radius ball need not vanish under proportional asymptotics (Dicker, 2016). Nevertheless, a fixed coordinate can be estimated consistently. The main difficulty in this regime is that the sample covariance is singular when $p > n$, and estimating the inverse-covariance direction associated with the target introduces an additional high-dimensional nuisance parameter. Therefore, the core question this paper addresses is what is the minimax rate of estimation of a fixed linear functional of β under this dense proportional regime.

When the design covariance is known, the coefficients of the linear model can be estimated consistently via the method of moments (Bradic et al., 2022), and they need to assume a sparse precision matrix when it is not known. Debiasing methods are another important type of approaches. Bellec

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and Zhang (2022) propose the degrees-of-freedom adjustment method which starts from a regularized regression estimator and corrects its bias using a precise analysis of proportional asymptotics. Bellec and Zhang (2023) extend debiasing to general convex penalties and establish asymptotic normality for most normalized directions. Celentano and Wainwright (2023) further develop debiasing for a population mean under missingness and confidence intervals for regression coefficients when both outcome and propensity estimators can be inconsistent. From a Bayesian perspective, Sáenz and Sur (2025) characterize fixed-dimensional posterior marginals under isotropic Gaussian designs and strongly log-concave product priors.

When the design covariance is unknown, the debiasing methods exploiting the sparsity of both the coefficient vector and the precision matrix are widely studied. The debiased Lasso methods (Zhang and Zhang, 2014; van de Geer et al., 2014) remove regularization bias through an approximately orthogonal score. Javanmard and Montanari (2014) construct an approximate inverse through constrained optimization, Bühlmann (2013) considers correcting the projection bias of ridge regression. These approaches impose different restrictions on regression sparsity, precision-direction sparsity, and the design. In particular, the method of Javanmard and Montanari (2014) does not require a sparse precision matrix, although it requires a sufficiently sparse regression vector. The analysis is later sharpened for Gaussian designs (Javanmard and Montanari, 2018). Similar approaches include post-double-selection inference (Belloni et al., 2014), decorrelated score tests (Ning and Liu, 2017), and double machine learning with orthogonal scores and cross-fitting (Chernozhukov et al., 2018). Their $\sqrt{n}$ -consistency results usually require suitable control of the nuisance estimation errors, often through a product remainder. Orthogonality reduces sensitivity to nuisance errors, but does not by itself make this remainder negligible when both nuisance estimators have nonvanishing errors under proportional asymptotics. Zhu and Bradic (2018a,b); Bradic et al. (2022) relaxed the sparsity assumption and develop tests that remain valid when β is dense, if the precision direction associated with the loading vector v is sparse. Cai and Guo (2017) establish corresponding minimax confidence-interval lengths and adaptation limits over sparse parameter spaces.

In the dense regime when both β and Σ^{-1} are dense, the source of covariance information and the inferential target become decisive. Celentano and Montanari (2024) propose correlation adjusted debiasing, which corrects bias caused by correlated errors in the outcome and covariate regressions. Their analysis permits inaccurate nuisance estimation, including overlapping samples. Other nonsparse linear-functional methods exploit additional spectral structure or informative initial estimators (Zhao et al., 2024). Li and Sur (2026) develop spectrum-aware debiasing for right-rotationally invariant designs and establish a spectral universality extension. Chen et al. (2024) is the closest work to the setting of our paper. They use higher-order moments to approximate the inverse covariance and estimate the resulting moments with higher-order U -statistics. Their construction gives coordinate consistency when $p \geq n$ without requiring a sparse precision matrix, but does not determine the optimal statistical rate. They later generalize the method of moments to the linear model with affine constrain (Sankaranarayanan et al., 2025) and the generalized linear model (Sawaya et al., 2026).

There are also a wide range of works studying the estimation of scalar functionals of linear model other than the linear functional. Dicker (2014) develop moment estimators of variance components, and Verzelen and Gassiat (2018) study adaptive estimation of signal-to-noise ratios with known or unknown covariance. Kong and Valiant (2018) show how polynomial approximation and moment estimation can recover prediction-related quantities using fewer observations than features. The dependence on covariance information in these problems motivates estimating a functional directly rather than first reconstructing the regression vector. Their targets and lower-bound regimes, however, differ from the fixed-coordinate proportional problem. For another functional, McGrath et al.

(2025) obtain $\sqrt{n}$ -consistency of an expected conditional covariance under isotropic independent-entry designs, despite inconsistent ridge nuisance estimators.

Despite the rich literature discussed above, it remains open (Maleki et al., 2026) what is the optimal statistical rate of estimating a fixed linear functional of the proportionally high-dimensional linear models when both the regression vector and the precision matrix are dense. This paper determines the minimax rate for this problem and characterizes the optimal estimator via the proportion γ and the design covariance matrix's condition number κ . Specifically, let ℓ and L be fixed positive lower and upper bounds of the covariance spectrum, and write $\kappa = L/\ell \geq 1$. We consider the design has the representation $X = Z\Sigma^{1/2}$, where the entries of Z are independent, identically distributed with standardized variance and bounded moments. The coefficient vector β has a bounded ℓ_2 -norm, and the independent Gaussian noise has an unknown bounded variance. Let $\mathcal{P}(n, p, \gamma, \kappa)$ be such model family, specified in Section 2, we prove

$$\inf_{\tilde{\beta}_v} \sup_{P \in \mathcal{P}(n, p, \gamma, \kappa)} \mathbb{E}_P(\tilde{\beta}_v - \beta_v)^2 \asymp n^{-\alpha(\kappa, \gamma)}. \quad (1.1)$$

The exponent $\alpha(\kappa, \gamma) \in (0, 1]$ has a closed-form expression (2.4) in terms of elliptic integrals. In specific, we have $\alpha(1, \gamma) = 1$ when $\gamma > 1$ and $\alpha(\kappa, \gamma) \in (0, 1)$ when both $\kappa > 1$ and $\gamma > 1$. Therefore, we can have the $\sqrt{n}$ -rate when design covariance is diagonal, which is known by, e.g., Bradic et al. (2022). However, the $\sqrt{n}$ -rate is not achievable under the general dense proportionally high-dimensional linear models, especially when $\kappa > 1$. When both κ and γ diverge, we show that $\alpha(\kappa, \gamma) \sim 4/(\sqrt{\kappa} \log \gamma)$; see Remark 3.7 for details. This asymptotic further shows the dimension-sample ratio γ only affects the rate logarithmically, while the covariance condition number κ affects the rate exponent in the order of $1/\sqrt{\kappa}$.

We propose the Bilinear Operator Calibration via Contour Aggregation (Bocca) estimator, which combines empirically calibrated resolvents through a complex contour integral. The calibration is motivated by the companion Stieltjes equation (Silverstein and Choi, 1995) and anisotropic local law (Knowles and Yin, 2017) for sample covariance matrices. A contour integral then combines the estimated resolvent field with an analytic filter. We show the rate of Bocca estimator is characterized by the bias terms determined by the approximation error of the filter on the covariance spectrum, while its growth on the observation contour controls the variance term. We identify a reflected elliptic-function as the filter which balances the bias-variance trade-off and attains the optimal rate.

Paper organization. Section 2 specifies the statistical models setup and notation related to the problem. Section 3 presents the Bocca estimator and its rate. Section 4 establishes the minimax lower bound.

2 Problem setup

Fix $0 < \ell < L$, $M, B_0, \bar{\sigma} > 0$, and set $\kappa = L/\ell$. For integers $p > n \geq 1$ and the compatible ratio $\gamma = p/n$, let $\mathcal{P}(n, p, \gamma, \kappa)$ be the family of joint laws of $(X, Y) \in \mathbb{R}^{n \times p} \times \mathbb{R}^n$ generated by

$$X = Z\Sigma^{1/2}, \quad Y = X\beta + \varepsilon, \quad \ell I_p \preceq \Sigma \preceq LI_p, \quad \|\beta\|_2 \leq B_0, \quad \text{and } 0 \leq \sigma \leq \bar{\sigma}. \quad (2.1)$$

Here Σ and β are deterministic parameters, the entries of Z are independent with an identical distribution satisfying $\mathbb{E}Z_{11} = 0$, $\mathbb{E}Z_{11}^2 = 1$, and $\mathbb{E}|Z_{11}|^r \leq M$ for every integer $r \geq 3$, and $\varepsilon \sim N(0, \sigma^2 I_n)$ is independent of Z . The constants $L, B_0, \bar{\sigma}, M$ are held fixed and are suppressed from the family notation. The covariance matrix Σ , regression coefficients β , and noise variance σ^2 are all unknown. Define $\mathcal{G}(n, p, \gamma, \kappa)$ as the subclass in which $Z_{11} \sim N(0, 1)$. Consider a sufficiently

large M , so that $\mathcal{G}(n, p, \gamma, \kappa) \subseteq \mathcal{P}(n, p, \gamma, \kappa)$. We will show the minimax lower bound is attained in the Gaussian design family $\mathcal{G}(n, p, \gamma, \kappa)$. All rate statements in this paper concern sequences of integers $p = p(n)$ for which

$$n \rightarrow \infty \text{ and } 1 < \liminf_{n \rightarrow \infty} \frac{p(n)}{n} \leq \limsup_{n \rightarrow \infty} \frac{p(n)}{n} < \infty. \quad (2.2)$$

The symbol γ always denotes the current ratio p/n . No convergence of p/n to a particular number is required. The number $\kappa = L/\ell$ specifies the eigenvalue bounds of the family; it need not be the condition number of each covariance matrix. As the diagonal covariance matrix case has been well studied (Bradic et al., 2022), in the rest of the paper, we focus on the general dense proportionally high-dimensional linear models when $\kappa > 1$.

Our target is $\beta_v = \langle v, \beta \rangle$ for a known deterministic vector $v \in \mathbb{R}^p$ with $\|v\|_2 = 1$; a nonunit loading is handled by rescaling. Write $\widehat{\Sigma} = X^\top X/n$ and $\widehat{S} = XX^\top/n$. The trace is unnormalized. We use $a \lesssim b$ when $a \leq Cb$, with C depending only on the fixed model constants and ratio bounds, and $a \asymp b$ for both inequalities.

2.1 Elliptic-integral notation

The incomplete elliptic integral of the first kind is

$$\mathbf{F}(\phi \mid m) = \int_0^\phi (1 - m \sin^2 t)^{-1/2} dt, \text{ for some } 0 < m < 1. \quad (2.3)$$

Its complete value is $\mathbf{K}(m) = \mathbf{F}(\pi/2 \mid m)$. Also $\cosh z = (e^z + e^{-z})/2$ and $\sinh z = (e^z - e^{-z})/2$. For the exponent, define

$$\alpha(\kappa, \gamma) = 1 - \frac{\mathbf{F}(\arcsin(\xi/\rho) \mid \rho^2)}{\mathbf{K}(\rho^2)}, \text{ where } \xi = \frac{\sqrt{\gamma} - 1}{\sqrt{\gamma} + 1} \text{ and } \rho = \frac{\sqrt{\gamma} - 1 + \kappa^{-1}}{\sqrt{\gamma} + 1 - \kappa^{-1}}. \quad (2.4)$$

Here $0 < \xi < \rho < 1$, so $0 < \alpha(\kappa, \gamma) < 1$.

3 The Bocca estimator

For the eigendecomposition $\Sigma = \sum_{j=1}^p \lambda_j v_j v_j^\top$, define $\nu = \sum_{j=1}^p (v^\top v_j)(v_j^\top \beta) \delta_{\lambda_j}$. Then the parameter of interest $\beta_v = \int \nu(d\lambda)$ and

$$F(\mu) = v^\top \Sigma (\Sigma - \mu I_p)^{-1} \beta = \int_{[\ell, L]} \frac{\lambda}{\lambda - \mu} \nu(d\lambda). \quad (3.1)$$

Given some holomorphic function g with $g(0) = 1$ and some contour Γ enclosing $[\ell, L] \cup \{0\}$, we consider the following filter

$$-\frac{1}{2\pi i} \oint_\Gamma \frac{1 - g(z)}{z} F(z) dz = \int \left[-\frac{1}{2\pi i} \oint_\Gamma \frac{1 - g(z)}{z} \frac{\lambda}{\lambda - z} dz \right] \nu(d\lambda) = \int_{[\ell, L]} (1 - g(\lambda)) \nu(d\lambda).$$

Throughout the paper, we always orient the contour integration counterclockwise. As our parameter of interest is $\beta_v = \int 1 \nu(d\lambda)$, to approximate it using the right hand side of the above equation, we need to find a g being small enough on the slit $[\ell, L]$, while $g(0) = 1$ removes the extraneous pole at zero. This is the key insight of our Bocca estimator to be proposed formally later.

We first construct an estimator for $F(\mu)$. Consider the Stieltjes transform

$$m(z) = \int_{\mathbb{R}} \frac{1}{x-z} \varrho(dx), \text{ where } \varrho = \frac{1}{p} \sum_{j=1}^p \delta_{\lambda_j}.$$

It is characterized in the upper half-plane by the equation

$$z = -\frac{1}{m(z)} + \frac{1}{n} \operatorname{tr}(\Sigma(I_p + m(z)\Sigma)^{-1}); \quad (3.2)$$

see [Silverstein and Choi \(1995, Eq \(1.3\)–\(1.4\)\)](#). We can utilize it to inverse $m(z)$ by

$$\zeta(\mu) = \mu - \frac{\mu}{n} \operatorname{tr}(\Sigma(\Sigma - \mu I_p)^{-1}). \quad (3.3)$$

The following proposition, whose proof is given in [Appendix D](#), shows when we can solve the inverse of $m(z)$ using $\zeta(\mu)$ on a certain region of the complex plane.

Proposition 3.1. For every $\mu \in \mathbb{C}$ with $|\mu - L| > \sqrt{\gamma}L$, the function m is analytic at $\zeta(\mu)$ and satisfies $m(\zeta(\mu)) = -1/\mu$.

For a deterministic $\delta > 0$, set

$$R_\delta = \sqrt{\gamma}(1 + \delta)L, \Gamma_\delta = \{z \in \mathbb{C} : |z - L| = R_\delta\}, \text{ and } \mathbb{D}_\delta = \{z \in \mathbb{C} : |z - L| < R_\delta\}.$$

The circle encloses $[\ell, L]$ and zero, and [Proposition 3.1](#) ensures $m(\zeta(\mu)) = -1/\mu$ for all μ on the contour Γ_δ , where we can define an estimator for $F(\mu)$.

We construct an estimator for $F(\mu)$ by using the anisotropic local law for sample covariance matrices ([Knowles and Yin, 2017](#)). They showed that

$$\frac{1}{n} X^\top (\hat{S} - zI_n)^{-1} X \approx m(z)\Sigma(I_p + m(z)\Sigma)^{-1},$$

Therefore, as $m(\zeta(\mu)) = -1/\mu$, we have $F(\mu) \approx n^{-1}(Xv)^\top (\hat{S} - \zeta(\mu)I_n)^{-1}Y$. Next is to estimate $\zeta(\mu)$. Define $\hat{m}(z) = n^{-1} \operatorname{tr}(\hat{S} - zI_n)^{-1}$. We next construct an estimator for $\zeta(\mu)$ by inverting $\hat{m}(z)$. Whenever it exists, define $\hat{\zeta} : \Gamma_\delta \rightarrow \mathbb{C}$ as the continuous function satisfying

$$\hat{m}(\hat{\zeta}(\mu)) = -1/\mu, \hat{\zeta}(\mu) \notin [\lambda_{\min}(\hat{S}), \lambda_{\max}(\hat{S})], \hat{\zeta}(L - R_\delta) < \lambda_{\min}(\hat{S}) \text{ and } \hat{m}'(\hat{\zeta}(\mu)) \neq 0, \quad (3.4)$$

for all $\mu \in \Gamma_\delta$. The derivative condition means that each selected root is simple. We set $\hat{\zeta}(\mu) = 0$ if no such function exists. The following proposition shows that such a function exists with high probability. Its proof is deferred to [Appendix D.4](#).

Proposition 3.2. Let $1/\log n \leq \delta \leq 2/\log n$. For every fixed $D > 0$ and all sufficiently large n , with probability at least $1 - Cn^{-D}$ there exists a unique continuous function $\hat{\zeta} : \Gamma_\delta \rightarrow \mathbb{C}$ satisfying [\(3.4\)](#) for all $\mu \in \Gamma_\delta$.

We discuss the concrete method to compute $\hat{\zeta}(\mu)$ in [Section 3.2](#). As the above proposition shows that $\hat{\zeta}(\mu)$ is well-defined with high probability, we can define the estimator for $F(\mu)$ as

$$\hat{F}(\mu) = \frac{1}{n} (Xv)^\top (\hat{S} - \hat{\zeta}(\mu)I_n)^{-1}Y. \quad (3.5)$$

If no such function exists, set $\hat{F}$ identically equal to zero.

For every deterministic function g , we define the Bilinear Operator Calibration via Contour Aggregation (*Bocca*) estimator as

$$\widehat{\beta}_v(g) = \text{clip}_{[-B_0, B_0]} \text{Re} \left[-\frac{1}{2\pi i} \oint_{\Gamma_\delta} \frac{1-g(z)}{z} \widehat{F}(z) \, dz \right]. \quad (3.6)$$

where $\text{clip}_{[-B_0, B_0]} A = \min\{\max\{A, B_0\}, -B_0\}$ is to ensure the estimator belongs to the model family that $\|\beta\|_2 \leq B_0$.

3.1 Bias–variance trade-off

Now we discuss how the function g determines the bias-variance trade-off of the Bocca estimator and how to construct an ideal g to achieve the optimal rate. To construct such g , we start with the harmonic potential u_δ satisfying

$$\Delta u_\delta = 0 \text{ in } \mathbb{D}_\delta \setminus [\ell, L], \quad u_\delta = 0 \text{ on } [\ell, L], \quad u_\delta = 1 \text{ on } \Gamma_\delta. \quad (3.7)$$

Define $\alpha_\delta = (u_\delta(0) - \delta)/(1 - \delta)$. We can prove that (see Lemma G.1 in the Appendix for details)

$$\alpha_\delta = \frac{u_\delta(0) - \delta}{1 - \delta} = \frac{\alpha(\kappa, \gamma(1 + \delta)^2) - \delta}{1 - \delta} = \alpha(\kappa, \gamma) + O(\delta). \quad (3.8)$$

Therefore, the key insight is that the optimal exponent $\alpha(\kappa, \gamma)$ for the estimation rate of β_v is determined by $u_0(0)$, the harmonic potential at zero.

We can actually have a closed form solution to u_δ . Put $k = \frac{R_\delta - L + \ell}{R_\delta + L - \ell}$ and $T_\delta(z) = \frac{R_\delta - L + z}{R_\delta + L - z}$. For $z \in \mathbb{D}_\delta \setminus [\ell, L]$, define $b_\delta(z)$ as

$$b_\delta(z) = \frac{2 \text{Re } T_\delta(z)}{\sqrt{(1 + |T_\delta(z)|^2 + |1 - T_\delta(z)|^2)(k^2 + |T_\delta(z)|^2 + |k^2 - T_\delta(z)|^2)}}. \quad (3.9)$$

Then $0 < b_\delta(z) < 1$, and

$$u_\delta(z) = 1 - \frac{\mathbf{F}(\arcsin b_\delta(z) \mid k^2)}{\mathbf{K}(k^2)} \text{ and } u_\delta(0) = 1 - \frac{\mathbf{F}\left(\arcsin \frac{R_\delta - L}{k(R_\delta + L)} \mid k^2\right)}{\mathbf{K}(k^2)}. \quad (3.10)$$

Let $\widetilde{u}_\delta$ be a local harmonic conjugate of u_δ , namely $u_\delta + i\widetilde{u}_\delta$ is holomorphic, normalized by $\widetilde{u}_\delta(0) = 0$, and define the complexification of u_δ as

$$q_\delta(z) = \frac{\pi \mathbf{K}(k^2)}{\mathbf{K}(1 - k^2)} (u_\delta(z) + i\widetilde{u}_\delta(z)). \quad (3.11)$$

The above results are derived from the slit-disk form of the classical elliptic rectangle construction (see Whittaker and Watson (1920, Sections 22.3, 22.12 and 22.35) and Section G.3 for details).

The follow theorem with its proof deferred to Appendix B.1 shows the statistical rate of the generic Bocca estimator.

Theorem 3.3. Let $1/\log n \leq \delta \leq 2/\log n$. Suppose g is deterministic, holomorphic on an open neighborhood of the closure of $\mathbb{D}_\delta$, and $g(0) = 1$. Define $\Lambda_\delta = \{z \in \mathbb{D}_\delta \setminus [\ell, L] : u_\delta(z) = \delta\}$ and

$$B(g) = \sup_{z \in \Lambda_\delta} |g(z)|^2, \quad V(g) = \sup_{z \in \Gamma_\delta} |g(z)|^2. \quad (3.12)$$

Then, for all sufficiently large n , we have

$$\sup_{P \in \mathcal{P}(n, p, \gamma, \kappa)} \mathbb{E}_P(\widehat{\beta}_v(g) - \beta_v)^2 \lesssim B(g) + \frac{V(g)}{n}. \quad (3.13)$$

The right-hand side of (3.13) explicitly shows how the bias-variance trade-off depends on g : the variance term is determined by the maximum value of g on the contour Γ_δ and the bias term is determined by the maximum value of g on Λ_δ which will shrink to the slit $[\ell, L]$ as δ goes to zero. The next theorem, whose proof is deferred in Appendix B.2, gives a constraint on these two values and gives the closed form of the ideal g to achieve the optimal rate.

Theorem 3.4. For sufficiently small δ , every function g holomorphic in $\{z \in \mathbb{D}_\delta \setminus [\ell, L] : \delta < u_\delta(z) < 1\}$, continuous on its two boundaries, and normalized by $g(0) = 1$ satisfies

$$B(g)^{1-\alpha_\delta} V(g)^{\alpha_\delta} \geq 1. \quad (3.14)$$

For every integer $K \geq 1$, the function

$$g_K(z) = \frac{\cosh(Kq_\delta(z))}{\cosh(Kq_\delta(0))} \quad (3.15)$$

has a holomorphic extension on a neighborhood of the closure of $\mathbb{D}_\delta$, satisfies $g_K(0) = 1$, and obeys

$$\begin{aligned} B(g_K) &\leq 4 \exp \left(-\frac{2\pi K(1-\delta)\mathbf{K}(k^2)}{\mathbf{K}(1-k^2)} \alpha_\delta \right), \\ V(g_K) &\leq 4 \exp \left(\frac{2\pi K(1-\delta)\mathbf{K}(k^2)}{\mathbf{K}(1-k^2)} (1-\alpha_\delta) \right). \end{aligned} \quad (3.16)$$

Consequently $1 \leq B(g_K)^{1-\alpha_\delta} V(g_K)^{\alpha_\delta} \leq 4$.

Remark 3.5. The above trade-off (3.14) is essentially the two-constants theorem (see Ransford (1995, Theorem 4.3.7)). The key insight why $u_0(0)$ implies the minimax exponent is because the extremal case of (3.14) is achieved by the harmonic measure by the two-constants theorem. In fact, the equality in (3.14) is achieved by $\exp(K(q_\delta(z) - q_\delta(0)))$. However, it has different values on the two sides of the slit $[\ell, L]$ so that it cannot be holomorphically extended to a neighborhood of the closure of $\mathbb{D}_\delta$. The hyperbolic cosine function in (3.15) symmetrizes the optimal exponential function, and makes it holomorphic on the whole disc. As is shown in Theorem 3.4, although g_K does not attain the equality in (3.14), it still achieves the constant upper bound of the left-hand side of (3.14), which is enough to achieve the optimal rate.

Combining the generic rate (3.13) and the bias-variance trade-off (3.14) already suggests the rate $n^{-\alpha_\delta}$ which converges to the minimax rate $n^{-\alpha(\kappa, \gamma)}$ as the margin δ shrinks. In particular, we can choose K in g_K to balance the two exponentials in (3.16). Following this idea, the theorem below gives the estimator to achieve the optimal rate.

Theorem 3.6. Choose $\delta \in [1/\log n, 2/\log n]$ and

$$K = \left\lfloor \frac{\mathbf{K}(1-k^2) \log n}{2\pi(1-\delta)\mathbf{K}(k^2)} \right\rfloor. \quad (3.17)$$

Then the Bocca estimator with g_K in (3.15) satisfies

$$\sup_{P \in \mathcal{P}(n, p, \gamma, \kappa)} \mathbb{E}_P(\hat{\beta}_v(g_K) - \beta_v)^2 \lesssim n^{-\alpha(\kappa, \gamma)}. \quad (3.18)$$

The proof of the theorem is in Appendix B.3. We will show in Theorem 4.1 that the above rate is minimax optimal.

Remark 3.7. The exponent $\alpha(\kappa, \gamma)$ is strictly decreasing in both $\kappa > 1$ and $\gamma > 1$. Therefore, increasing either the covariance condition number or the dimension ratio slows the minimax rate. When both κ and γ diverge, we can show that

$$\alpha(\kappa, \gamma) \sim \frac{4}{\sqrt{\kappa} \log \gamma}. \quad (3.19)$$

See Proposition G.3 in the appendix for details. This shows that the optimal rate is less sensitive to the growth of the proportion γ than the covariance condition number κ .

3.2 Computation

We now discuss the computation of the Bocca estimator. One of the nice properties of $\hat{\beta}_v(g)$ in (3.6) is that it is linear in v before the clipping. Therefore, we can compute the estimator before specifying the loading v .

Contour discretization. We compute the contour integral in (3.6) by discretizing the contour circle Γ_δ into M equally spaced points. Define $t_j = \pi + \frac{2\pi j}{M}$, $\mu_j = L + R_\delta e^{it_j}$, for $j = 0, \dots, M$. The nodes traverse Γ_δ counterclockwise, starting at $\mu_0 = L - R_\delta < 0$, and $\mu_M = \mu_0$. Compute the eigendecomposition of $\hat{S} = XX^\top/n = U \text{diag}(s_1, \dots, s_n)U^\top$. Then $\hat{m}(z) = n^{-1} \sum_{i=1}^n (s_i - z)^{-1}$.

Computing $\hat{\zeta}$ by Newton's method. As $\hat{m}(\hat{\zeta}(\mu)) = -1/\mu$, for each node μ_j , we can solve z from the equation

$$\frac{1}{n} \sum_{i=1}^n \frac{1}{s_i - z} + \frac{1}{\mu_j} = 0. \quad (3.20)$$

We start with the real node $\mu_0 = L - R_\delta < 0$, as $s_1 + \mu_0 \leq \hat{\zeta}(\mu_0) \leq s_1 + \frac{\mu_0}{n}$, we can solve $z_0 = \hat{\zeta}(\mu_0)$ by bisection on the interval $[s_1 + \mu_0, s_1 + \frac{\mu_0}{n}]$. We then continue to compute the next node iteratively applying the Newton's method to solve (3.20). Given $z_j = \hat{\zeta}(\mu_j)$ is computed, we describe how to compute $z_{j+1} = \hat{\zeta}(\mu_{j+1})$. Differentiate $\hat{m}(\hat{\zeta}(\mu)) = -1/\mu$ to obtain

$$\hat{\zeta}'(\mu) = \left[\mu^2 \frac{1}{n} \sum_{i=1}^n (s_i - \hat{\zeta}(\mu))^{-2} \right]^{-1}. \quad (3.21)$$

We then apply the Newton's method

$$z_{j+1}^{(r+1)} = z_{j+1}^{(r)} - \frac{n^{-1} \sum_{i=1}^n (s_i - z_{j+1}^{(r)})^{-1} + \mu_{j+1}^{-1}}{n^{-1} \sum_{i=1}^n (s_i - z_{j+1}^{(r)})^{-2}}, \text{ where } z_{j+1}^{(0)} = z_j + \frac{\mu_{j+1} - \mu_j}{\mu_j^2 n^{-1} \sum_{i=1}^n (s_i - z_j)^{-2}}.$$

We will stop the iteration when achieved the desired tolerances and denote the resulting value as z_{j+1} .

Deterministic kernel values. For the specific choice of $g = g_K$ in (3.15), we compute its value on the contour $\mu(t) = L + R_\delta e^{it}$ without computing $q_\delta(z)$ in its definition. For $t \notin 2\pi\mathbb{Z}$, set

$$y(t) = \mathbf{F} \left(\arctan \frac{\cot(t/2)}{k} \middle| 1 - k^2 \right).$$

The reflected kernel can then be evaluated directly as

$$g_K(\mu(t)) = \cosh \left(\frac{\pi K}{\mathbf{K}(1 - k^2)} (\mathbf{K}(k^2) - iy(t)) \right) / \cosh \left(\frac{\pi K \mathbf{K}(k^2)}{\mathbf{K}(1 - k^2)} u_\delta(0) \right), \quad (3.22)$$

where $u_\delta(0)$ is given in (3.10). At $t \in 2\pi\mathbb{Z}$, use either one-sided limit. The two limits agree because $y(t)$ changes by $2\mathbf{K}(1-k^2)$ and K is an integer. Thus all kernel evaluations require only real elliptic integrals and a complex hyperbolic cosine. Those formulas are verified in Appendix G.

Bocca estimator computation for all loadings. Compute the real weights

$$w_i = -\frac{1}{M} \operatorname{Re} \sum_{j=0}^{M-1} \frac{(\mu_j - L)(1 - g(\mu_j))}{\mu_j(s_i - z_j)}, \text{ for } i = 1, \dots, n, \quad (3.23)$$

and the whole coefficient vector estimator

$$\tilde{\beta}_M(g) = \frac{1}{n} X^\top \left[U \left(\operatorname{diag}(w_1, \dots, w_n) (U^\top Y) \right) \right].$$

For a given v , we compute the Bocca estimator by $\hat{\beta}_{v,M}(g) = \operatorname{clip}_{[-B_0, B_0]}(v^\top \tilde{\beta}_M(g))$.

For the g_K in (3.15) with the conjugation symmetry, we have

$$\mu_{M-j} = \overline{\mu_j}, \hat{\zeta}(\mu_{M-j}) = \overline{\hat{\zeta}(\mu_j)}, g_K(\mu_{M-j}) = \overline{g_K(\mu_j)}, \text{ for } 1 \leq j < M.$$

Thus one may compute grids on a single semicircle of Γ_δ and obtain the remaining values by conjugation.

4 Information-theoretic lower bound

In this section, we establish the lower bound which matches the rate of the Bocca estimator in Theorem 3.6.

Theorem 4.1. Given any fixed v with unit length $\|v\|_2 = 1$, let $\tilde{\beta}_v$ be any estimator for β_v . Then we have

$$\inf_{\tilde{\beta}_v} \sup_{P \in \mathcal{G}(n, p, \gamma, \kappa)} \mathbb{E}_P(\tilde{\beta}_v - \beta_v)^2 \gtrsim n^{-\alpha(\kappa, \gamma)}. \quad (4.1)$$

Consequently, the same lower bound holds with $\mathcal{P}(n, p, \gamma, \kappa)$.

Proof sketch. We first take $\ell = \kappa^{-1}$, $L = 1$, and $v = e_1$. The general bounds and loading follow from a fixed scaling and orthogonal transformation; see Appendix C. In this normalized construction, two priors induce data laws P_0, P_1 such that

$$\operatorname{TV}(P_0, P_1) \leq \frac{1}{3}, \quad \beta_1 = b_j \text{ under prior } j, \quad |b_1 - b_0| \geq cn^{-\alpha(\kappa, \gamma)/2}.$$

The construction begins with two $(p-1)$ -covariate regressions having the same design marginal. Integrating out their random coefficients conditionally on that design gives Gaussian response laws. The difference of their conditional variances is a rational transform of two positive atomic measures ν_0, ν_1 . Endpoint weights ensure the final covariance bounds; a weighted spectral integral controls the expected conditional Kullback–Leibler divergence. A common Gaussian transformation converts the difference in the preliminary noise variances into a difference in the final first coefficients. The two-hypothesis testing bound then proves the result. Appendix F gives the complete construction.

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References

- ABRAMOWITZ, M. and STEGUN, I. A. (eds.) (1964). *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. No. 55 in Applied Mathematics Series, National Bureau of Standards, Washington, DC.
- AHLFORS, L. V. (1979). *Complex Analysis*. 3rd ed. McGraw–Hill, New York.
- BANDEIRA, A. S., BOEDIHARDJO, M. and VAN HANDEL, R. (2023). Matrix concentration inequalities and free probability. *Inventiones Mathematicae* **234** 419–487.
- BAYER, C. and TEICHMANN, J. (2006). The proof of Tchakaloff’s theorem. *Proceedings of the American Mathematical Society* **134** 3035–3040.
- BELLEÇ, P. C. and ZHANG, C.-H. (2022). De-biasing the Lasso with degrees-of-freedom adjustment. *Bernoulli* **28** 713–743.
- BELLEÇ, P. C. and ZHANG, C.-H. (2023). Debiasing convex regularized estimators and interval estimation in linear models. *Annals of Statistics* **51** 391–436.
- BELLONI, A., CHERNOZHUKOV, V. and HANSEN, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *The Review of Economic Studies* **81** 608–650.
- BOUCHERON, S., BOUSQUET, O., LUGOSI, G. and MASSART, P. (2005). Moment inequalities for functions of independent random variables. *Annals of Probability* **33** 514–560.
- BRADIC, J., FAN, J. and ZHU, Y. (2022). Testability of high-dimensional linear models with nonsparse structures. *Annals of Statistics* **50** 615–639.
- BRAILOVSKAYA, T. and VAN HANDEL, R. (2024). Universality and sharp matrix concentration inequalities. *Geometric and Functional Analysis* **34** 1734–1838.
- BÜHLMANN, P. (2013). Statistical significance in high-dimensional linear models. *Bernoulli* **19** 1212–1242.
- CAI, T. T. and GUO, Z. (2017). Confidence intervals for high-dimensional linear regression: Minimax rates and adaptivity. *Annals of Statistics* **45** 615–646.
- CELENTANO, M. and MONTANARI, A. (2024). Correlation adjusted debiased Lasso: Debiasing the Lasso with inaccurate covariate model. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **86** 1455–1482.
- CELENTANO, M. and WAINWRIGHT, M. J. (2023). Challenges of the inconsistency regime: Novel debiasing methods for missing data models. ArXiv preprint arXiv:2309.01362.
- CHEN, X., LIU, L. and MUKHERJEE, R. (2024). Method-of-moments inference for GLMs and doubly robust functionals under proportional asymptotics. ArXiv preprint arXiv:2408.06103. Version 3, May 2025.
- CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., DUFLO, E., HANSEN, C., NEWEY, W. and ROBINS, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21** C1–C68.

- DICKER, L. H. (2014). Variance estimation in high-dimensional linear models. *Biometrika* **101** 269–284.
- DICKER, L. H. (2016). Ridge regression and asymptotic minimax estimation over spheres of growing dimension. *Bernoulli* **22** 1–37.
- HE, Y., KNOWLES, A. and ROSENTHAL, R. (2018). Isotropic self-consistent equations for mean-field random matrices. *Probability Theory and Related Fields* **171** 203–249.
- JAVANMARD, A. and MONTANARI, A. (2014). Confidence intervals and hypothesis testing for high-dimensional regression. *Journal of Machine Learning Research* **15** 2869–2909.
- JAVANMARD, A. and MONTANARI, A. (2018). Debiasing the Lasso: Optimal sample size for Gaussian designs. *Annals of Statistics* **46** 2593–2622.
- KNOWLES, A. and YIN, J. (2017). Anisotropic local laws for random matrices. *Probability Theory and Related Fields* **169** 257–352.
- KONG, W. and VALIANT, G. (2018). Estimating learnability in the sublinear data regime. In *Advances in Neural Information Processing Systems*, vol. 31.
- LI, Y. and SUR, P. (2026). Spectrum-aware debiasing: A modern inference framework with applications to principal components regression. *Annals of Statistics* **54** 745–770. Preprint: arXiv:2309.07810.
- MALEKI, A., SEN, S., BALAKRISHNA, S., ZUBER, V., GAO, C., DUDEJA, R., THRAMPOULIDIS, C., ZHANG, A., SU, W., KLUSOWSKI, J. M., LOH, P.-L. and SHOJAIE, A. (2026). High-dimensional statistics: Reflections on progress and open problems. *arXiv preprint arXiv:2605.05076* .
- MCGRATH, S., MUKHERJEE, D., MUKHERJEE, R. and WANG, Z. J. (2025). Optimal nuisance function tuning for estimating a doubly robust functional under proportional asymptotics. In *Advances in Neural Information Processing Systems*, vol. 38.
- NING, Y. and LIU, H. (2017). A general theory of hypothesis tests and confidence regions for sparse high dimensional models. *Annals of Statistics* **45** 158–195.
- RANSFORD, T. (1995). *Potential theory in the complex plane*. No. 28 in London Mathematical Society Student Texts, Cambridge university press.
- SÁENZ, M. and SUR, P. (2025). Characterizing finite-dimensional posterior marginals in high-dimensional GLMs via leave-one-out. ArXiv preprint arXiv:2601.00091.
- SANKARANARAYANAN, M., HRYTSENKO, Y., ROTTER, J. I., SOFER, T. and MUKHERJEE, R. (2025). Asymptotic inference for constrained regression.
- SAWAYA, K., UEMATSU, Y. and IMAIZUMI, M. (2026). Moment-based adjustments of statistical inference in high-dimensional generalized linear models. *Electronic Journal of Statistics* **20** 4083–4133. Preprint: arXiv:2305.17731.
- SILVERSTEIN, J. W. and CHOI, S.-I. (1995). Analysis of the limiting spectral distribution of large dimensional random matrices. *Journal of Multivariate Analysis* **54** 295–309.

- TREFETHEN, L. N. (2013). *Approximation Theory and Approximation Practice*. Society for Industrial and Applied Mathematics, Philadelphia.
- VAN DE GEER, S., BÜHLMANN, P., RITOV, Y. and DEZEURE, R. (2014). On asymptotically optimal confidence regions and tests for high-dimensional models. *Annals of Statistics* **42** 1166–1202.
- VERZELEN, N. and GASSIAT, E. (2018). Adaptive estimation of high-dimensional signal-to-noise ratios. *Bernoulli* **24** 3683–3710.
- WHITTAKER, E. T. and WATSON, G. N. (1920). *A Course of Modern Analysis*. 3rd ed. Cambridge University Press, Cambridge. Chapter XXII, “The Jacobian elliptic functions,” 491–535.
- YANG, F. (2024). Linear spectral statistics of eigenvectors of anisotropic sample covariance matrices. *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques* **60** 2767–2812.
- ZHANG, C.-H. and ZHANG, S. S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *Journal of the Royal Statistical Society, Series B* **76** 217–242.
- ZHAO, J., ZHOU, Y. and LIU, Y. (2024). Estimation of linear functionals in high-dimensional linear models: From sparsity to nonsparsity. *Journal of the American Statistical Association* **119** 1579–1591.
- ZHU, Y. and BRADIC, J. (2018a). Linear hypothesis testing in dense high-dimensional linear models. *Journal of the American Statistical Association* **113** 1583–1600.
- ZHU, Y. and BRADIC, J. (2018b). Significance testing in non-sparse high-dimensional linear models. *Electronic Journal of Statistics* **12** 3312–3364.
- ZWIERNIK, P., UHLER, C. and RICHARDS, D. (2017). Maximum likelihood estimation for linear Gaussian covariance models. *Journal of the Royal Statistical Society, Series B* **79** 1269–1292.

Appendix

A Notation and preliminaries for the proofs

Norms, matrices, and distances. For a complex random variable T and $r \geq 1$, write $\|T\|_{L^r} = (\mathbb{E}|T|^r)^{1/r}$. If ψ is a deterministic function on a contour, then $\|\psi\|_{\infty, \Gamma_\delta} = \sup_{z \in \Gamma_\delta} |\psi(z)|$. We use $\|A\|_{\text{op}}$ for the Euclidean operator norm, $\|A\|_F = (\text{tr } A^* A)^{1/2}$ for the Frobenius norm, and $A^* = \overline{A}^\top$. The relation $A \preceq B$ means $x^*(B - A)x \geq 0$ for all vectors x . The Hausdorff distance between two sets A, B is $\text{dist}_H(A, B) = \sup_{a \in A, b \in B} \|a - b\|_2$. For probability laws P, Q , we write

$$\text{TV}(P, Q) = \sup_E |P(E) - Q(E)|, \quad D_{\text{KL}}(P\|Q) = \int \log(dP/dQ) dP,$$

with divergence $+\infty$ when P is not absolutely continuous with respect to Q . For complex variables, $\text{Cov}(T, S) = \mathbb{E}[(T - \mathbb{E}T)(S - \mathbb{E}S)]$ is the bilinear covariance; the Hermitian covariance is $\text{Cov}(T, \bar{S})$ and $\text{Var}(T) = \mathbb{E}|T - \mathbb{E}T|^2$. For a real signed measure η , $\|\eta\|_{\text{TV}} = |\eta|(\mathbb{R})$ denotes its total variation mass. These conventions differ by a factor two when $\eta = P - Q$. The variation of a real function is denoted by $\text{Var}_{\mathbb{R}}(f)$, to distinguish it from both probability distance and statistical variance.

Matrix blocks and entry derivatives. The first p coordinates of $\mathbb{C}^{p+n}$ form the feature block, and the last n coordinates form the observation block. Let $P_1 = \text{diag}(I_p, 0)$ and $P_2 = \text{diag}(0, I_n)$. In block-matrix expressions, a vector in $\mathbb{C}^p$, including s_k below, is identified with its embedding in the first block. In products with $p \times p$ matrices it retains its original dimension. Write $\mathbf{e}_i = (0, e_i) \in \mathbb{R}^{p+n}$ for the i th observation-block vector, $s_k = \Sigma^{1/2} e_k \in \mathbb{R}^p$, and $x_{ki} = Z_{ik}/\sqrt{n}$. For a differentiable matrix statistic, ∂_{ki} means differentiation with respect to the real variable x_{ki} , with every other entry held fixed. A spectral derivative, such as $G'(z)$, differentiates its complex argument instead. For smooth functions on $\mathbb{C}$, the complex derivatives are $\partial_z = (\partial_x - i\partial_y)/2$ and $\partial_{\bar{z}} = (\partial_x + i\partial_y)/2$; $dA = dx dy$ is planar area measure. In a bilinear matrix entry, $A_{vw} = v^\top A w$; this notation does not include complex conjugation. An observation index, as in G_{iw} , abbreviates $G_{\mathbf{e}_i, w}$.

Set $\mathcal{X} = \Sigma^{1/2} Z^\top / \sqrt{n}$. The generalized resolvent and its deterministic comparison matrix are

$$G(z) = \begin{pmatrix} -I_p & \mathcal{X} \\ \mathcal{X}^\top & -zI_n \end{pmatrix}^{-1}, \quad \Pi(z) = \begin{pmatrix} -(I_p + m(z)\Sigma)^{-1} & 0 \\ 0 & m(z)I_n \end{pmatrix}. \quad (\text{A.1})$$

Their diagonal blocks are indexed by 11 and 22. In particular, $G_{22}(z) = (\hat{S} - zI_n)^{-1}$ and $G_{11}(z) = X^\top (\hat{S} - zI_n)^{-1} X / n - I_p$. Smooth versions, needed when a real spectral argument falls near a sample eigenvalue on an exceptional event, are constructed in Lemma D.2.

Two recurrent filter quantities. For the integer degree K , define the spectral error bound and the contour amplification by

$$\mathbf{b}_K = e^{-Kq_\delta(0)}, \quad \mathbf{v}_K = \exp \left(K \left[\frac{\pi \mathbf{K}(k^2)}{\mathbf{K}(1 - k^2)} - q_\delta(0) \right] \right). \quad (\text{A.2})$$

The bounds in (A.2) are used in the scalar and maximum-coordinate proofs. The modulus $k = (R_\delta - L + \ell)/(R_\delta + L - \ell)$ is the one used in Section 3.1. The geometric lemma gives $\sup_{[\ell, L]} |g_K| \leq 2\mathbf{b}_K$ and $\sup_{\Gamma_\delta} |(1 - g_K(z))/z| \leq C\mathbf{v}_K$.

Moment and derivative inequalities. For $1 \leq r \leq s$, Hölder's inequality gives $\|T\|_{L^r} \leq \|T\|_{L^s}$; this is the moment-monotonicity inequality also called Lyapunov's inequality. For a smooth compactly supported function f on $\mathbb{R}$, the Sobolev norm used below is $\|f\|_{W^{j,1}} = \sum_{a=0}^j \int_{\mathbb{R}} |f^{(a)}(x)| dx$. If f is holomorphic on a neighborhood of $|z - z_0| \leq r$, Cauchy's integral formula for derivatives gives

$$f^{(j)}(z_0) = \frac{j!}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{j+1}} dz, \quad |f^{(j)}(z_0)| \leq j! r^{-j} \sup_{|z-z_0|=r} |f(z)|. \quad (\text{A.3})$$

We use this form of Cauchy's derivative estimate, including its explicit power of r^{-1} ; see [Ahlfors \(1979, Chapter 4, Section 2.3, Eq \(24\)\)](#).

B Proofs of upper bounds

B.1 The generic boundary-norm bound

Proof of Theorem 3.3. The measure in (3.1) satisfies $\|\nu\|_{\text{TV}} \leq B_0$. By the assumed full-disk analyticity and $g(0) = 1$, the quotient $(1 - g(z))/z$ is holomorphic at zero, with value $-g'(0)$. Fubini's theorem and the residue theorem ([Ahlfors, 1979, Chapter 4, Section 5.1, Theorem 17](#)) give

$$-\frac{1}{2\pi i} \oint_{\Gamma_\delta} \frac{1 - g(z)}{z} F(z) dz = \int_{[\ell, L]} (1 - g(\lambda)) \nu(d\lambda).$$

The only possible pole of the inner integrand is at λ , with residue $-(1 - g(\lambda))$; it is removable if this residue vanishes. Compact separation of the contour and the interval justifies Fubini's theorem. The population error is therefore $-v^\top g(\Sigma)\beta$. Lemma G.1 and the maximum-modulus principle ([Ahlfors, 1979, Chapter 4, Section 3.4, Theorems 12 and 12'](#)) imply that its modulus is at most $B_0 \sqrt{B(g)}$.

Also $V(g) \geq |g(0)|^2 = 1$. The ratio assumption gives $\inf_{z \in \Gamma_\delta} |z| = L(\sqrt{\gamma}(1 + \delta) - 1) \geq c > 0$ uniformly for all sufficiently large n . Hence

$$\sup_{z \in \Gamma_\delta} \left| \frac{1 - g(z)}{z} \right|^2 \leq C(1 + \sqrt{V(g)})^2 \leq C'V(g).$$

Let $\mathcal{A}$ be the X -measurable event in Lemma E.1, with moment order two and failure exponent $D = 2$. In particular $P(\mathcal{A}^c) \leq Cn^{-2}$, the empirical inverse exists on $\mathcal{A}$, and

$$\mathbb{E}_P \left[\mathbf{1}_{\mathcal{A}} \left| \frac{1}{2\pi i} \oint_{\Gamma_\delta} \frac{1 - g(z)}{z} (\hat{F}(z) - F(z)) dz \right|^2 \right] \leq \frac{CV(g)}{n}.$$

Taking the real part and clipping contract error to the real target $\beta_v \in [-B_0, B_0]$. On $\mathcal{A}$, the squared error is consequently at most twice the squared stochastic contribution plus $2B_0^2 B(g)$. On $\mathcal{A}^c$ it is at most $4B_0^2$, including the zero fallback. Thus the risk is at most $2B_0^2 B(g) + CV(g)/n + Cn^{-2}$. Since $V(g) \geq 1$, the last term is absorbed into $CV(g)/n$. All bounds are uniform over the stated finite-dimensional family. $\square$

B.2 Two constants and extension through the spectrum

Proof of Theorem 3.4. We first compare the logarithmic boundary norms by the maximum principle, then verify holomorphic extension of the reflected function, and finally bound its two norms.

The logarithmic boundary constraint. For $z \in \mathbb{D}_\delta \setminus [\ell, L]$ with $u_\delta(z) > \delta$, define

$$\mathfrak{h}(z) = \left(1 - \frac{u_\delta(z) - \delta}{1 - \delta}\right) \log B(g) + \frac{u_\delta(z) - \delta}{1 - \delta} \log V(g).$$

If either $B(g)$ or $V(g)$ is zero, first use its value plus $\varepsilon > 0$ and then let $\varepsilon \downarrow 0$ in the comparison below; $g(0) = 1$ rules out that case. Thus both logarithms are finite. This harmonic function equals $\log B(g)$ on Λ_δ and $\log V(g)$ on Γ_δ . The function $\log |g|^2 - \mathfrak{h}$ is subharmonic and has nonpositive boundary upper limits. The subharmonic maximum principle therefore gives $\log |g(0)|^2 \leq \mathfrak{h}(0)$; see Ahlfors (1979, Chapter 6, Section 4.1). Since $g(0) = 1$, we have $0 \leq (1 - \alpha_\delta) \log B(g) + \alpha_\delta \log V(g)$, which proves (3.14). At zeros, $\log |g| = -\infty$ is interpreted in the usual subharmonic sense; alternatively apply the comparison to $\max(\log |g|, -M)$ and then let $M \rightarrow \infty$.

Extension of the reflected function. Lemma G.1 gives an increment $2\pi i$ for q_δ around the slit. Thus integer K makes $\cosh(Kq_\delta)$ single-valued on $\mathbb{D}_\delta \setminus [\ell, L]$. Its two traces on the open slit satisfy

$$q_{\delta,-}(x) = -q_{\delta,+}(x) \pmod{2\pi i}, \quad g_{K,-}(x) = g_{K,+}(x), \quad \ell < x < L.$$

Fix $x \in (\ell, L)$ and choose $r < \min(x - \ell, L - x)$. On the disk $|z - x| < r$, assign the common trace to g_K on the real diameter. This defines a continuous function. For a triangle crossing that diameter, apply Cauchy's theorem to the portions at imaginary distance at least ϵ above and below it, and then let $\epsilon \downarrow 0$. The integrals along their horizontal cuts converge to opposite integrals of the same continuous trace. They cancel, so the integral along the triangle is zero. Morera's theorem yields a holomorphic function on the disk; see Ahlfors (1979, Chapter 4, Section 2.3, p. 122). Varying x gives a holomorphic extension to $\mathbb{D}_\delta \setminus \{\ell, L\}$.

For every fixed K , the potential satisfies $0 \leq \operatorname{Re} q_\delta \leq \pi \mathbf{K}(k^2)/\mathbf{K}(1 - k^2)$ on the slit disk. The identity $|\cosh(x + iy)| \leq \cosh x$ therefore bounds g_K in punctured neighborhoods of both endpoints, including the values just assigned on the open slit. Consequently $(z - \ell)g_K(z) \rightarrow 0$ and $(z - L)g_K(z) \rightarrow 0$. The removable-singularity theorem in Ahlfors (1979, Chapter 4, Section 3.1, Theorem 7) extends g_K through both endpoints.

Schwarz reflection is the following extension principle: a holomorphic function continuous up to a real interval and real-valued there extends to the other side by $f(z) = \overline{f(\bar{z})}$; see Ahlfors (1979, Chapter 4, Section 6.5, Theorem 24). On the outer circle $\operatorname{Re} q_\delta$ is constant. Subtract that constant, multiply by i , and take a local fractional-linear coordinate sending the circle to a real interval. The principle then continues the local determination of q_δ across the circle. The period-invariant function g_K has consistent continuations on overlaps. Compactness of the circle supplies an exterior neighborhood. Since $q_\delta(0) > 0$ is real, its denominator is nonzero and $g_K(0) = 1$.

The boundary estimates. The real part of q_δ is $\delta \pi \mathbf{K}(k^2)/\mathbf{K}(1 - k^2)$ on Λ_δ and $\pi \mathbf{K}(k^2)/\mathbf{K}(1 - k^2)$ on Γ_δ . For $x \geq 0$, $|\cosh(x + iy)| \leq e^x$ and $\cosh x \geq e^x/2$. Substituting these bounds in (3.15) gives

$$\sup_{\Lambda_\delta} |g_K| \leq 2 \exp \left(-K \left[q_\delta(0) - \frac{\delta \pi \mathbf{K}(k^2)}{\mathbf{K}(1 - k^2)} \right] \right), \quad \sup_{\Gamma_\delta} |g_K| \leq 2 \mathfrak{v}_K.$$

By the definition of α_δ , the two exponents are respectively $-K(1 - \delta) \pi \mathbf{K}(k^2) \alpha_\delta / \mathbf{K}(1 - k^2)$ and $K(1 - \delta) \pi \mathbf{K}(k^2) (1 - \alpha_\delta) / \mathbf{K}(1 - k^2)$. Squaring proves (3.16). Their product with powers $1 - \alpha_\delta$ and α_δ is at most four. The reverse bound follows from (3.14). $\square$

B.3 The optimized scalar rate

Proof of Theorem 3.6. We first bound the two envelopes at the stated integer degree and then compare the resulting exponent with its limiting value. The definition (3.17) gives

$$\frac{\mathbf{K}(1 - k^2) \log n}{2\pi(1 - \delta)\mathbf{K}(k^2)} - 1 \leq K \leq \frac{\mathbf{K}(1 - k^2) \log n}{2\pi(1 - \delta)\mathbf{K}(k^2)}.$$

Using (3.16), we obtain

$$B(g_K) \leq 4 \exp\left(\frac{2\pi(1 - \delta)\mathbf{K}(k^2)\alpha_\delta}{\mathbf{K}(1 - k^2)}\right) n^{-\alpha_\delta}, \quad \frac{V(g_K)}{n} \leq 4n^{-\alpha_\delta}.$$

Lemma G.1 bounds the displayed prefactor uniformly over the fixed parameter class. Theorem 3.3 therefore gives $\sup_{P \in \mathcal{P}(n, p, \gamma, \kappa)} \mathbb{E}_P(\hat{\beta}_v(g_K) - \beta_v)^2 \leq Cn^{-\alpha_\delta}$. Moreover, (3.8) and $\delta \log n \leq 2$ imply

$$n^{-\alpha_\delta} = n^{-\alpha(\kappa, \gamma)} \exp((\alpha(\kappa, \gamma) - \alpha_\delta) \log n) \leq e^{2C} n^{-\alpha(\kappa, \gamma)}.$$

This proves (3.18). □

C Proofs of lower bounds

C.1 The scalar minimax lower bound

Proof of Theorem 4.1. We construct preliminary Gaussian experiments with covariance eigenvalues in $[\kappa^{-1}, 1]$, then scale their data to the fixed L in (2.1) and rotate to the prescribed loading. The preliminary experiments use their own coefficient and noise bounds; the transformation below places their priors inside the final family.

The first coordinate with eigenvalues in $[\kappa^{-1}, 1]$. Consider $v = e_1$, so the parameter of interest is β_1 . Lemma F.2 constructs two priors $\mathbb{Q}_0, \mathbb{Q}_1$ supported on Gaussian regression parameters satisfying $\kappa^{-1}I_p \preceq \Sigma \preceq I_p$, $\|\beta\|_2 \leq C\sqrt{\tau}$, and $\sigma^2 \in [(1 + \kappa^{-1})/3, (1 + \kappa^{-1})/2]$. Let P_0, P_1 be their induced data laws. The lemma gives

$$\text{TV}(P_0, P_1) \leq \frac{1}{3}, \quad \mathbb{Q}_j(\beta_1 = b_j) = 1, \quad |b_1 - b_0| \geq cn^{-\alpha(\kappa, p/n)/2}.$$

Relabel the priors so that $b_0 < b_1$. For any estimator T of β_1 , use the test $\varphi = \mathbf{1}_{\{T > (b_0 + b_1)/2\}}$. Under $\mathbb{Q}_0$, the event $\varphi = 1$ implies $|T - \beta_1| \geq (b_1 - b_0)/2$; under $\mathbb{Q}_1$, the analogous implication holds on $\varphi = 0$. Therefore

$$\sup_{\Sigma, \beta, \sigma} \mathbb{E}(T - \beta_1)^2 \geq \frac{(b_1 - b_0)^2}{8} (P_0(\varphi = 1) + P_1(\varphi = 0)) \geq \frac{(b_1 - b_0)^2}{12}.$$

The last step uses the identity $\inf_\varphi [P_0(\varphi = 1) + P_1(\varphi = 0)] = 1 - \text{TV}(P_0, P_1)$.

General eigenvalue bounds and loading. Denote a parameter in the preceding construction by $(\Sigma^\circ, \beta^\circ, \sigma^\circ)$ and its data by (X°, Y°) . Choose an orthogonal matrix O satisfying $Oe_1 = v$. For a fixed $c_0 > 0$, transform the data by

$$X = \sqrt{L} X^\circ O^\top, \quad Y = c_0 Y^\circ.$$

This gives the parameter map

$$\Sigma = L O \Sigma^\circ O^\top, \quad \beta = \frac{c_0}{\sqrt{L}} O \beta^\circ, \quad \sigma = c_0 \sigma^\circ.$$

We have $\ell I_p \preceq \Sigma \preceq L I_p$, because $L/\kappa = \ell$. Choose c_0 so that $c_0 \sqrt{(1 + \kappa^{-1})/2} \leq \bar{\sigma}$, and then reduce the fixed τ in the prior so that $(c_0/\sqrt{L}) C \sqrt{\tau} \leq B_0$. Each transformed fixed-parameter law belongs to $\mathcal{G}(n, p, \gamma, \kappa)$. The data transformation is invertible and common to the two hypotheses, so it preserves their total variation distance. Moreover,

$$\beta_v = v^\top \beta = \frac{c_0}{\sqrt{L}} e_1^\top \beta^\circ.$$

The target separation is multiplied by the fixed positive number $c_0/\sqrt{L}$. The same testing argument proves (4.1) on $\mathcal{G}(n, p, \gamma, \kappa)$. The inclusion in $\mathcal{P}(n, p, \gamma, \kappa)$ and Theorem 3.6 then give the two scalar minimax comparisons stated in the theorem. $\square$

D Calibration, spectral control, and the empirical inverse

In this section, we study $\zeta(\mu)$ and $\hat{\zeta}(\mu)$ and prove Propositions 3.1 and 3.2.

D.1 The deterministic calibration and quantitative separation

Proof of Proposition 3.1. For $|\mu - L| > \sqrt{\gamma}L$ and $\lambda \in [\ell, L]$, the triangle inequality gives

$$|\lambda - \mu| \geq |\mu - L| - (L - \lambda) > \sqrt{\gamma}L - (L - \lambda) \geq \sqrt{\gamma}\lambda.$$

Consequently $n^{-1} \sum_j \lambda_j^2 / |\lambda_j - \mu|^2 < 1$. The condition also excludes zero and all population eigenvalues. Substitution of (3.3) shows that $s = -1/\mu$ solves (3.2) at $z = \zeta(\mu)$, and

$$\operatorname{Im} \zeta(\mu) = \operatorname{Im} \mu \left(1 - \frac{1}{n} \sum_j \frac{\lambda_j^2}{|\lambda_j - \mu|^2} \right).$$

For $\operatorname{Im} \mu > 0$, both $\zeta(\mu)$ and $-1/\mu$ belong to the upper half-plane. Uniqueness of its Stieltjes solution (Silverstein and Choi, 1995, equations (1.3)–(1.4)) gives the identity. Conjugation gives the lower-half-plane case.

For real μ in the stated set,

$$\zeta'(\mu) = 1 - \frac{1}{n} \sum_j \frac{\lambda_j^2}{(\lambda_j - \mu)^2} > 0.$$

At $s = -1/\mu$, the derivative of the inverse representation in (3.2) is $\mu^2 \zeta'(\mu) > 0$. Also $s \neq 0$ and $-1/s = \mu$ is outside the population spectrum. Theorem 4.2 of Silverstein and Choi (1995) therefore gives $\zeta(\mu) \notin \operatorname{supp} \varrho$ and $m(\zeta(\mu)) = s$. A Stieltjes transform is analytic off its support, completing the proof. $\square$

The following additional estimates are used in the stochastic proofs. They are stated separately because Proposition 3.1 asserts only the qualitative identity.

Lemma D.1 (Quantitative separation on the observation contour). For $|\mu - L| \geq \sqrt{\gamma}(1 + \delta)L$, we have

$$\operatorname{Re} \zeta'(\mu) \geq 1 - (1 + \delta)^{-2} > 0. \quad (\text{D.1})$$

The map ζ is injective on this set. For all sufficiently small $\delta > 0$ and every $\mu \in \Gamma_\delta$,

$$\operatorname{dist}(\zeta(\mu), \operatorname{supp} \varrho) \geq c\delta^2, \quad |m'(\zeta(\mu))| \geq c. \quad (\text{D.2})$$

On sufficiently small neighborhoods of these image points, fixed-order derivatives of m and the deterministic resolvent expressions are bounded by fixed powers of δ^{-1} .

Proof. The same triangle inequality, now with a margin, gives $n^{-1} \sum_j \lambda_j^2 / |\lambda_j - \mu|^2 \leq (1 + \delta)^{-2}$. Differentiation proves (D.1). Direct subtraction gives

$$\zeta(\mu) - \zeta(\nu) = (\mu - \nu) \left(1 - \frac{1}{n} \sum_j \frac{\lambda_j^2}{(\lambda_j - \mu)(\lambda_j - \nu)} \right). \quad (\text{D.3})$$

The Cauchy–Schwarz inequality bounds the modulus of the sum by $(1 + \delta)^{-2}$. Thus $|\zeta(\mu) - \zeta(\nu)| \geq (1 - (1 + \delta)^{-2})|\mu - \nu|$, proving injectivity.

For a point $\mu \in \Gamma_\delta$, a disk of radius $c_1\delta$ about μ , with sufficiently small fixed c_1 , lies in $|\nu - L| > \sqrt{\gamma}(1 + \delta/2)L$. Equation (D.3) bounds the distance of the image of its boundary from $\zeta(\mu)$ below by $c_2\delta^2$. The argument principle, or Rouché’s theorem applied to $\zeta(\nu) - \zeta(\mu)$, implies that the image contains the disk of radius $c_2\delta^2$ about $\zeta(\mu)$. Throughout this image Proposition 3.1 identifies m with $-1/\nu$. A real image point has a real preimage: otherwise conjugation would give two preimages in the strict exterior, contradicting (D.3). The real case of the proposition therefore excludes every point of $\operatorname{supp} \varrho$ from this image disk. This proves the first estimate in (D.2).

Differentiating the identified transform gives

$$m'(\zeta(\mu)) = \frac{1}{\mu^2 \zeta'(\mu)}. \quad (\text{D.4})$$

On the circle, $|\mu|$ is bounded above and away from zero and $|\zeta'| \leq 2$, yielding its lower bound. On the image disks, the Stieltjes integral gives $|m^{(j)}(z)| \leq j! \operatorname{dist}(z, \operatorname{supp} \varrho)^{-j-1}$. The population factors have denominators $\lambda - \nu$ separated from zero on the corresponding preimages. These estimates and Cauchy’s integral formula for derivatives give the remaining fixed-power bounds. The complex-analytic theorems used here are in Ahlfors (1979, Chapter 4, Sections 5.1–5.2). $\square$

D.2 A spectral event and preliminary moments

The complex neighborhoods used below have the explicit form

$$\left\{ z \in \mathbb{C} : \inf_{\mu \in \Gamma_\delta} |z - \zeta(\mu)| < c_1 \delta^b \right\},$$

with fixed $b \geq 2$ and sufficiently small $c_1 > 0$. Lemma D.1 places them away from $\operatorname{supp} \varrho$. When a smaller neighborhood is required for derivatives, c_1 is decreased or b increased; no new statistical tuning parameter is introduced.

We next define the smooth extension of the Green matrix in (A.1). Write

$$\mathcal{H} = \begin{pmatrix} 0 & \mathcal{X} \\ \mathcal{X}^\top & 0 \end{pmatrix}, P_1 = \operatorname{diag}(I_p, 0), P_2 = \operatorname{diag}(0, I_n),$$

where $\mathcal{X} = \Sigma^{1/2} Z^\top / \sqrt{n}$. The matrices P_1 and P_2 project onto the first p and last n coordinates, respectively. For each of finitely many neighborhoods covering $\zeta(\Gamma_\delta)$, choose an even function $\chi \in C^\infty(\mathbb{R})$ that equals one on a neighborhood of the nonzero spectrum of $\mathcal{H}$ on the spectral-separation event, and vanishes near zero, near the real poles t satisfying $t^2 = z$ for z in that neighborhood, and outside a fixed compact interval. The transition regions can be chosen to have widths that are fixed powers of δ . For each argument z in that neighborhood, define smooth scalar functions on $\mathbb{R}$ by

$$f_{0,z}(t) = \frac{\chi(t)}{t^2 - z}, \quad f_{1,z}(t) = t f_{0,z}(t), \quad f_{2,z}(t) = t^2 f_{0,z}(t),$$

assigning value zero on the neighborhoods where χ vanishes. The smooth version of $G(z)$ in (A.1) is

$$\tilde{G}(z) = P_1 f_{2,z}(\mathcal{H}) P_1 - P_1 + P_2 f_{0,z}(\mathcal{H}) P_2 + P_1 f_{1,z}(\mathcal{H}) P_2 + P_2 f_{1,z}(\mathcal{H}) P_1. \quad (\text{D.5})$$

Let $\tilde{m}(z) = \frac{1}{n} \text{tr}(P_2 \tilde{G}(z) P_2)$. On the spectral-separation event, $\tilde{G}(z) = G(z)$, $\tilde{m}(z) = \hat{m}(z)$, for every z in the corresponding neighborhood. Outside this event, $\tilde{G}$ and $\tilde{m}$ remain smooth functions of the entries of Z , with every fixed derivative bounded by a fixed power of δ^{-1} .

The next lemma records the quantitative input needed by the moment expansion. Its proof checks both the comparison parameters and the passage through the companion gap. A centered bilinear form of a complex matrix is interpreted by its complex absolute moments; real and imaginary parts are treated separately when a real-valued concentration inequality is used.

Lemma D.2 (Preliminary Green-function bounds). Let $1/\log n \leq \delta \leq 2/\log n$. On neighborhoods of $\zeta(\Gamma_\delta)$ whose widths are fixed powers of δ , let $\tilde{G}(z)$ and $\tilde{m}(z)$ be the smooth extensions defined in (D.5). For every fixed $D > 0$, there exists an X -measurable event $\mathcal{A}_n$ such that $P(\mathcal{A}_n^c) \leq C_D n^{-D}$. On $\mathcal{A}_n$, $\tilde{G}(z) = G(z)$, $\tilde{m}(z) = \hat{m}(z)$, $cI_n \preceq \hat{S} \preceq CI_n$. Moreover, the spectrum of $\hat{S}$ is separated from the corresponding spectral parameters by a fixed power of δ . In specific, $\|(\hat{S} - zI_n)^{-1}\|_{\text{op}} \leq C\delta^{-2}$.

For every fixed $r \geq 2$, every sufficiently small fixed $\epsilon > 0$, and deterministic vectors $v, w \in \mathbb{C}^{p+n}$, we have

$$\begin{aligned} \|v^\top \{\tilde{G}(z) - \Pi(z)\} w\|_{L^r} &\leq C_{r,\epsilon} n^{-1/2+\epsilon} \delta^{-C_{r,\epsilon}}, \\ \|\tilde{m}(z) - \mathbb{E}\tilde{m}(z)\|_{L^r} &\leq C_{r,\epsilon} n^{-1+\epsilon} \delta^{-C_{r,\epsilon}}. \end{aligned} \quad (\text{D.6})$$

The same bounds hold for every fixed-order derivative with respect to z , with an additional fixed power of δ^{-1} .

Remark D.3. The operator comparisons used here are Brailovskaya and van Handel (2024, Theorems 2.6 and 2.10) and Bandeira et al. (2023, Theorems 2.1 and 2.8); the moment step uses Boucheron et al. (2005, Theorem 2). We compute their matrix parameters below. The spectral distance is allowed to decrease as a power of $1/\log n$, so the our proof records every occurrence of the inverse distance, by giving sharper mean and moment statements.

Proof. We establish spectral separation and smooth deterministic equivalents first, and then use entry replacement to bound fluctuations.

Independent summands and spectral comparison. Since $\mathcal{X}^\top \mathcal{X} = \hat{S}$, consider the Hermitian dilation $\mathcal{H} = \begin{pmatrix} 0 & \mathcal{X} \\ \mathcal{X}^\top & 0 \end{pmatrix}$. It is a sum over the independent entries $Z_{ik}/\sqrt{n}$ of the deterministic matrices $s_k \mathbf{e}_i^\top + \mathbf{e}_i s_k^\top$, where s_k and $\mathbf{e}_i$ are the block vectors defined in Appendix A. First truncate the unnormalized entries at n^{ϵ_0} , center, and standardize. For any desired fixed probability and moment errors, the all-orders envelope makes the total truncation probability and the centering and variance

corrections smaller than those errors. The number of entries is $O(n^2)$. The proof exponent ϵ_0 is chosen after the fixed moment order; it is not an estimator parameter.

In the notation of [Brailovskaya and van Handel \(2024\)](#), the variance parameter is bounded, the maximal bilinear variance is $O(n^{-1})$, and the largest summand norm is $O(n^{-1/2+\epsilon_0})$. Indeed summing the squared deterministic entry matrices divided by n gives the diagonal blocks Σ and $(\text{tr } \Sigma/n)I_n$. The bilinear bound follows from $\sum_k s_k s_k^\top = \Sigma$. For the matching Gaussian series, the covariance operator of its vectorized entries has norm at most $2L/n$. Thus the product denoted by $\tilde{v}^4$ in [Bandeira et al. \(2023\)](#) is $O(n^{-1})$.

Theorem 2.6 of [Brailovskaya and van Handel \(2024\)](#), with concentration parameter a sufficiently large multiple of $\log n$, followed by Theorem 2.1 of [Bandeira et al. \(2023\)](#), confines the spectrum within

$$C \left(n^{-1/6+\epsilon_0/3} (\log n)^{2/3} + n^{-1/4} (\log n)^{3/4} + n^{-1/2+\epsilon_0} \log n \right)$$

of the matching deterministic free spectrum, with any fixed polynomial failure probability. This error is smaller than every fixed power of δ . The hypotheses of these results concern independent Hermitian summands and their listed parameters; they impose no regularity condition on individual sample-covariance edges.

To identify the deterministic free resolvent, expand it in inverse powers of the spectral parameter outside a fixed large disk. The semicircular moment formula pairs entries in noncrossing pairs. Isolating the pair containing the first random entry gives the matrix self-consistency equation; the independent top–bottom entry coefficients make its diagonal blocks diagonal. In the covariance eigenbasis the resulting equations are $g_j(z) = (z - \lambda_j g_0(z))^{-1}$ and $g_0(z) = (z - n^{-1} \sum_j \lambda_j g_j(z))^{-1}$. The convergent expansions establish these identities initially for large arguments; analytic continuation and the Stieltjes sign determine their solution elsewhere. Substitution $g_0(z) = -zm(z^2)$ gives (3.2). The trace in the deterministic free model is faithful, so its support identifies the operator spectrum. Only these deterministic equations are diagonalized; a rotation of the non-Gaussian independent entries is never made.

Excluding an additional small companion eigenvalue. The ordinary dilation has $p - n$ structural zero modes, so confinement alone would not exclude a small additional singular value. Apply the same two comparison theorems to

$$\begin{pmatrix} I_p & \mathcal{X} \\ \mathcal{X}^\top & -I_n \end{pmatrix}.$$

Its eigenvalues are $\pm \sqrt{1 + s_j(\mathcal{X})^2}$ and $p - n$ copies of $+1$. The deterministic bottom covariance is bounded below by the identity-covariance bottom model multiplied by ℓ , whose lower edge is $(\sqrt{\gamma} - 1)^2$. The shifted deterministic model therefore has a fixed gap around -1 . Spectral confinement of the shifted model excludes small sample singular values. The deterministic shift does not alter any summand or variance parameter. This proves $cI_n \preceq \hat{S} \preceq CI_n$ with the stated probability. Together with Lemma D.1, it provides the needed resolvent neighborhoods, including calibrated values equal to zero.

Smooth matrix functions and their expectations. We use the generalized resolvent (A.1). Its top block equals $X^\top (\hat{S} - zI_n)^{-1} X/n - I_p$, and its bottom block is $(\hat{S} - zI_n)^{-1}$. The following construction specifies the smooth versions used on exceptional spectral events, including a spectral argument equal to zero.

Fix a small neighborhood of one deterministic spectral argument. The nonzero spectrum of the ordinary dilation is separated from all square roots of arguments in this neighborhood. Choose an even C^∞ function χ of the real variable which equals one on a smaller neighborhood of that nonzero spectrum, vanishes near zero and outside a fixed bounded set, and also vanishes near every such

real pole. The fixed positive gap at zero and the separation in Lemma D.1 allow the transition widths to be fixed powers of δ . Convolution of an indicator of a spectral neighborhood with a fixed smooth mollifier constructs such a function with $\|\chi^{(j)}\|_\infty \leq C_j \delta^{-C_j}$; this bound is independent of the number of spectral intervals. A finite cover of the spectral-argument curve suffices.

The smooth version of $G(z)$ in (D.5) is

$$\tilde{G}(z) = P_1 f_{2,z}(\mathcal{H}) P_1 - P_1 + P_2 f_{0,z}(\mathcal{H}) P_2 + P_1 f_{1,z}(\mathcal{H}) P_2 + P_2 f_{1,z}(\mathcal{H}) P_1.$$

In the expectation and cumulant calculations below, G denotes this smooth version; it equals the exact generalized resolvent on the spectral event. Algebraic resolvent identities and their entry derivatives below are first applied to the exact matrix on this event and on its one-entry perturbations. When taking expectations of the smooth versions, the resulting discrepancy is $O(n^{-D})$ for any prescribed fixed D , by their polynomial derivative bounds and the arbitrarily high probability exponent. Such terms are included in the stated remainders. Its normalized bottom trace is denoted by $\hat{m}$ in those calculations. In particular, its top nullspace value is $-I$ and its bottom block has no zero mode. Each scalar function is analytic in z and has every fixed Sobolev norm bounded by a fixed power of δ^{-1} . This also bounds fixed matrix-entry derivatives: write the scalar function by its Fourier integral, differentiate the matrix exponential using

$$\left. \frac{d}{ds} e^{it(A+sE)} \right|_{s=0} = it \int_0^1 e^{it(1-u)A} E e^{ituA} du,$$

iterate, and integrate the resulting bound against its Fourier transform. Integrating by parts in the scalar transform bounds the necessary Fourier moments by finitely many L^1 derivative norms. Thus these bounds also hold off the spectral event.

The operator-valued smooth comparison in Brailovskaya and van Handel (2024, Theorem 2.10) bounds the non-Gaussian/Gaussian expectation difference by $C(R\sigma^2 + R^3 \log(p+n))\|f\|_{W^{5,1}}$, in that theorem's notation. With the summand parameters above it is $O(n^{-1/2+\epsilon_0} \delta^{-C})$. The matching Gaussian/free comparison follows from the matrix-valued resolvent bound in Bandeira et al. (2023, Theorem 2.8). Lemma H.2 integrates that bound and gives $O(n^{-1}\|f\|_{W^{6,1}})$ for these compactly supported functions. Applying these estimates to the four block terms proves the preliminary mean estimate.

Resampling bounds. Write a normalized entry as x_{ki} , and replace it by an independent copy with difference d . On the spectral event, the rank-two resolvent identity and its second-order expansion give, with $s = s_k$,

$$|G'_{vw} - G_{vw}| \leq C|d|(|G_{vs}G_{iw}| + |G_{vi}G_{sw}|) + C|d|^2 \delta^{-C}(|G_{vs}G_{sw}| + |G_{vi}G_{iw}|).$$

All entries on the right can be taken from the unperturbed matrix; the remaining rank-two inverse is uniformly bounded because $|d| \leq Cn^{-1/2+\epsilon_0}$ is smaller than the spectral margin. Squaring and summing over (k, i) , using $\sum_k s_k s_k^\top = \Sigma$, bounds the discrete squared gradient by $Cn^{-1+4\epsilon_0} \delta^{-C}$. The first-order products have a bounded summed square before the factor $\max d^2$; the second-order same-block products have at most one additional factor $n+p$ before $\max d^4$.

For the normalized trace, differentiation gives $\partial_{ki} \hat{m} = -2(GP_2G)_{s_k i}/n$, where P_2 is the observation-block projection from Appendix A. The summed squared first derivatives are at most $C\delta^{-C}/n$, using the Frobenius bound on GP_2G . Second-order trace coefficients are at most $C\delta^{-C}/n$. Its discrete squared gradient is therefore $Cn^{-2+4\epsilon_0} \delta^{-C}$. The smooth versions have polynomial bounds on the complementary spectral events. Hölder's inequality and an arbitrarily small exceptional probability make their contribution negligible. All one-entry perturbations and their intervening segments are

covered by the rank-two perturbation bound; a finite polynomial union over replacements does not change the conclusion.

Apply [Boucheron et al. \(2005, Theorem 2\)](#) to the positive and negative parts of each centered real statistic, and then to its imaginary part. The theorem uses the conditional expectation of the squared replacement differences, which is bounded by the same estimates. Its moment conclusion yields fluctuations of sizes $n^{-1/2+2\epsilon_0}\delta^{-C_r}$ and $n^{-1+2\epsilon_0}\delta^{-C_r}$. Combining with the preliminary mean comparison and choosing ϵ_0 sufficiently small proves [\(D.6\)](#). The derivative bound in [\(A.3\)](#) supplies fixed-order spectral derivatives. The refinement is proved next and does not enter this preliminary argument. $\square$

D.3 Refining the mean and constructing the sample inverse

Lemma D.4 (Refined mean). On the same smooth neighborhoods, for every sufficiently small fixed $\epsilon > 0$,

$$|\mathbb{E}\hat{m} - m| + \|\mathbb{E}G_{11} - \Pi_{11}\|_{\text{op}} \leq C_\epsilon n^{-1+\epsilon}\delta^{-C_\epsilon}. \quad (\text{D.7})$$

Together with [Lemma D.2](#), this gives the trace approximation used in [Lemma D.5](#).

Proof. Work in the original independent-coordinate basis. Fix a spectral argument z , write $m = m(z)$, and put $D = I_p + m\Sigma$, $t = D^{-1}v$, and $c_k = t^\top s_k$. For the exact generalized resolvent, the top block equation in [\(A.1\)](#) gives

$$(G - \Pi)_{vw} = \sum_{k,i} c_k x_{ki} G_{iw} + m G_{\Sigma t, w}. \quad (\text{D.8})$$

For a normalized entry, the expansion of $\mathbb{E}[xf(x)]$ starts with the variance term $n^{-1}\mathbb{E}f'(x)$. The next terms are $\chi_3\mathbb{E}f''(x)/(2n^{3/2})$ and $\chi_4\mathbb{E}f'''(x)/(6n^2)$, where χ_j is the unnormalized entry cumulant. A finite expansion with its remainder is given in [Lemma E.2](#). The derivative rule is $\partial_{ki}G = -G(s_k e_i^\top + e_i s_k^\top)G$.

The second cumulant differentiates G_{iw} into $-G_{is_k}G_{iw} - G_{ii}G_{s_k w}$. Summing the second term cancels the final term in [\(D.8\)](#), up to the empirical trace error. Hence

$$\mathbb{E}(G - \Pi)_{vw} = -\mathbb{E}((\hat{m} - m)G_{\Sigma t, w}) - \frac{1}{n}\mathbb{E}(GP_2G)_{\Sigma t, w} + \text{higher cumulants}. \quad (\text{D.9})$$

The only potentially large third-cumulant self derivative is seen from

$$\partial_{ki}^2 G_{iw} = 2G_{is_k}^2 G_{iw} + 2G_{ii}G_{s_k s_k} G_{iw} + 4G_{ii}G_{is_k} G_{s_k w}.$$

In the middle term replace the two diagonal factors by their deterministic equivalents. These do not depend on i . The identity $\sum_i G_{iw} = \sqrt{n} G_{\bar{e}, w}$, where $\bar{e} = n^{-1/2} \sum_i e_i$ is a deterministic bottom vector, and [Lemma D.2](#) save a factor $\sqrt{n}$ over an absolute row sum. A replacement error has two coarse fluctuations and has the same saving. The remaining terms have either more mixed-block factors or the bounded contraction $\sum_k |c_k G_{s_k w}| \leq C\delta^{-C}$. Thus the third cumulant is $O(n^{-1+C_\epsilon}\delta^{-C})$, without an assumption of a symmetric entry law. The fourth cumulant with no mixed-block factors retains this endpoint contraction and is $O(n^{-1}\delta^{-C})$. For cumulant order five, four derivatives of the original mixed-block entry leave at least one mixed-block factor in every term. Even the bound $\sum_{k,i} |c_k| \leq Cn\sqrt{p}$, combined with its coefficient $n^{-5/2}$ and the mixed-entry bound $n^{-1/2+\epsilon}\delta^{-C}$, gives $O(n^{-3/2+\epsilon}\delta^{-C})$. For order six and higher, the coefficient $n^{-j/2}$ and the same sum already give $O(n^{-3/2}\delta^{-C})$. All other factors have bounded operator norms up to powers of δ^{-1} . [Lemma E.2](#), with a sufficiently large fixed order, gives a smaller remainder.

Let $d_m = \mathbb{E}\widehat{m}(z) - m(z)$. The fluctuation bound for the trace and the coarse bilinear estimate in the first term of (D.9) show, uniformly in deterministic unit v, w ,

$$\mathbb{E}(G - \Pi)_{vw} = d_m v^\top \Sigma D^{-2} w + O(n^{-1+C\epsilon}\delta^{-C}).$$

For the original matrices, $n^{-1} \text{tr}(G_{11} - \Pi_{11}) = z(\widehat{m} - m)$ holds exactly. The smooth matrices agree with them on an event whose complement has arbitrarily small inverse-polynomial probability. Since the smooth terms have polynomial bounds, their expectations satisfy

$$n^{-1} \text{tr}(\mathbb{E}G_{11} - \Pi_{11}) = z d_m + O(n^{-D})$$

for any prescribed fixed D . Therefore, we have

$$\left(z - \frac{1}{n} \text{tr}(\Sigma D^{-2})\right) d_m = O(n^{-1+C\epsilon}\delta^{-C}).$$

The coefficient equals $-m^{-1}(1 - m^2 \text{tr}(\Sigma^2 D^{-2})/n)$, whose modulus is at least $c\delta$ by the deterministic stability estimate. This argument remains valid when $z = 0$; no division by z is made. Reducing the preliminary proof exponent establishes (D.7). $\square$

Lemma D.5 (Quantitative control of the empirical inverse). Let $1/\log n \leq \delta \leq 2/\log n$. For every fixed $D > 0$, there is an event $\mathcal{A}$, measurable in X , with $P(\mathcal{A}^c) \leq C_D n^{-D}$, on which the continuous inverse in (3.4) exists and

$$\sup_{\mu \in \Gamma_\delta} |\widehat{\zeta}(\mu) - \zeta(\mu)| \leq C n^{-3/4} \delta^{-c}, \quad \sup_{\mu \in \Gamma_\delta} \|(\widehat{S} - \widehat{\zeta}(\mu) I_n)^{-1}\|_{\text{op}} \leq C \delta^{-c}. \quad (\text{D.10})$$

Here $c < \infty$ is fixed, $c_0 I_n \preceq \widehat{S} \preceq C_0 I_n$ on $\mathcal{A}$, and the inverse is locally analytic and satisfies $\widehat{\zeta}(\bar{\mu}) = \overline{\widehat{\zeta}(\mu)}$ on the circle.

Proof. We obtain uniform trace approximation, isolate one root near every deterministic root, and then verify the real initialization and global continuity.

Lemmas D.2 and D.4, a polynomial-size net on slightly smaller spectral neighborhoods from Lemma D.1, and sufficiently high fixed moments give

$$\sup_z |\widehat{m}(z) - m(z)| \leq C n^{-3/4} \delta^{-c}$$

with failure probability $C_D n^{-D}$. The supremum is over these neighborhoods, not over the entire complex plane. On the spectral event, derivatives in z are bounded by fixed powers of δ^{-1} ; a mesh n^{-A} with a fixed sufficiently large A therefore controls interpolation. The moment order and failure exponent are chosen after the mesh size. The event includes the spectral gap and upper norm from Lemma D.2.

Fix $\mu \in \Gamma_\delta$. By (D.4) and the polynomial bound on m'' , a circle of radius $r = c_1 \delta^b$ about $\zeta(\mu)$, for a sufficiently large fixed b , is inside the common resolvent set and satisfies

$$|m(z) + 1/\mu| \geq c_2 r \quad \text{when } |z - \zeta(\mu)| = r.$$

Taylor's theorem compared with $m'(\zeta(\mu))(z - \zeta(\mu))$ shows that $m(z) + 1/\mu$ has exactly one zero there, counted with multiplicity. The empirical perturbation is smaller than $c_2 r$ for large n , so Rouché's theorem (Ahlfors, 1979, Chapter 4, Section 5.2, corollary to Theorem 18) gives exactly one empirical zero in this disk, also counted with multiplicity. It is therefore simple. The same Taylor bound at that zero yields displacement at most $C n^{-3/4} \delta^{-c}$, uniformly in μ .

The displacement is $o(r)$. By the analytic implicit-function theorem, each selected root continues locally in μ . In a sufficiently small neighborhood of a given contour point, this continuation stays inside both the original isolating disk and the disk about the nearby deterministic root. The latter has exactly one zero, so the two selections coincide. This proves continuity on the circle. At the repeated endpoint of a parametrization the deterministic disk is the same and contains the same unique zero; hence the selection is a function on the closed circle.

At $\mu_0 = L - R_\delta < 0$, the deterministic root $\zeta(\mu_0)$ is below $\min \text{supp } \varrho$. To see this, follow the real map from $\mu = -\infty$, where $\zeta(\mu) \rightarrow -\infty$; its derivative remains positive and its image remains outside $\text{supp } \varrho$ by Proposition 3.1. The quantitative gap and sample spectral confinement keep its isolating disk below $\lambda_{\min}(\widehat{S})$ for large n . The disk and its equation are conjugation invariant, so its unique root is real. It is the unique solution below the smallest sample eigenvalue, since $\widehat{m}$ increases there from zero to infinity. Thus it has the required initial value.

Spectral separation and the displacement bound imply the resolvent estimate in (D.10). Conjugating the root construction and using uniqueness gives the stated symmetry. All events involve only X , with uniform constants over the model family. $\square$

D.4 Existence and uniqueness of the empirical inverse

Proof of Proposition 3.2. Lemma D.5 proves existence. To prove uniqueness, the value at $L - R_\delta$ is unique by strict monotonicity of $\widehat{m}$ below the smallest eigenvalue. The set where two selections agree is closed by continuity. It is also open in Γ_δ , since $\widehat{m}' \neq 0$ gives a locally unique inverse at each common value. This set is nonempty, and Γ_δ is connected, so the two selections agree everywhere. This establishes uniqueness of the continuous selection on the circle. $\square$

E Moments of the integrated calibrated field

This appendix proves the stochastic estimate shared by the scalar and vector results. Pointwise Green-function bounds are used only to control remainders. The leading terms are integrated through their two-point covariance kernels; this distinction prevents a leading power of $\log n$.

E.1 Cumulants and a finite integration-by-parts expansion

For a real variable Z with all moments finite, its j th cumulant is

$$\chi_j = (-i)^j \frac{d^j}{dt^j} \log \mathbb{E} e^{itZ} \Big|_{t=0}.$$

The logarithm is defined near zero, where the characteristic function is nonzero. This definition uses a characteristic function, not an assumed finite moment-generating function. Equivalently, the cumulants are determined recursively by

$$\mathbb{E} Z^{r+1} = \sum_{j=0}^r \binom{r}{j} \chi_{j+1} \mathbb{E} Z^{r-j}.$$

For a centered standardized entry, $\chi_1 = 0$, $\chi_2 = 1$, $\chi_3 = \mathbb{E} Z^3$, and $\chi_4 = \mathbb{E} Z^4 - 3$. Higher cumulants measure the additional terms beyond Gaussian integration by parts. We use the notation of He et al. (2018, Lemma 2.4); a sample-covariance formulation appears in Yang (2024, Lemma 3.7).

Lemma E.1 (Fixed moments of the calibrated field). For every fixed $r \geq 1$, every deterministic unit loading v , and every deterministic continuous weight ψ on Γ_δ , there is an X -measurable event $\mathcal{A}$ with $\mathbb{P}(\mathcal{A}^c) \leq C_D n^{-D}$ for any fixed D , such that

$$\mathbb{E} \left[\mathbf{1}_{\mathcal{A}} \left| \frac{1}{2\pi i} \oint_{\Gamma_\delta} \psi(z) (\widehat{F}(z) - F(z)) dz \right|^r \right] \leq C_r n^{-r/2} \|\psi\|_{\infty, \Gamma_\delta}^r. \quad (\text{E.1})$$

We prove Lemma E.1 in Appendix E.3 using the cumulant recursion and integrated-kernel estimate below.

Lemma E.2 (Cumulant expansion). Let Z be centered with variance one, and set $x = Z/\sqrt{n}$. Fix $r \geq 2$. Suppose $f \in C^r(\mathbb{R}; \mathbb{C})$, $\|f^{(r)}\|_\infty < \infty$, and $\mathbb{E}|Z|^{r+1} < \infty$. Then we have

$$\mathbb{E}[xf(x)] = \sum_{j=2}^r \frac{\chi_j}{(j-1)! n^{j/2}} \mathbb{E} f^{(j-1)}(x) + \mathcal{R},$$

where

$$|\mathcal{R}| \leq n^{-(r+1)/2} \|f^{(r)}\|_\infty \left(\frac{\mathbb{E}|Z|^{r+1}}{r!} + \sum_{j=2}^r \frac{|\chi_j| \mathbb{E}|Z|^{r+1-j}}{(j-1)!(r+1-j)!} \right).$$

Remark E.3. This is the standard cumulant expansion, specialized to entries of variance $1/n$ and a bounded-derivative remainder. The remainder in He et al. (2018, Lemma 2.4) instead separates a tail term and a local derivative bound; Yang (2024, Lemma 3.7) uses the same separation for sample covariance matrices. The bounded form above follows directly by Taylor's theorem, as shown next. For Gaussian Z , only χ_2 is nonzero, and the untruncated identity is Gaussian integration by parts, $\mathbb{E}[xf(x)] = n^{-1} \mathbb{E} f'(x)$, under the usual integrability conditions. As we need to have the uniform control of the Green-function derivatives on the logarithmic contour, for the self-containedness, we still present its proof below.

Proof. Expand $f(x)$ at zero through degree $r-1$. The remainder in $\mathbb{E}[xf(x)]$ is at most $\|f^{(r)}\|_\infty \mathbb{E}|x|^{r+1}/r!$. In the j th term on the right, expand $f^{(j-1)}(x)$ through degree $r-j$. Its remainder, after multiplication by the cumulant coefficient, is bounded by $|\chi_j| n^{-j/2} \|f^{(r)}\|_\infty \mathbb{E}|x|^{r+1-j}/((j-1)!(r+1-j)!)$. The polynomial coefficients agree by the moment-cumulant recursion above. Substituting $x = Z/\sqrt{n}$ gives the claimed bound.

For a fixed Green-function product, differentiating one entry inserts the rank-two matrix $\Delta_{ki} = s_k \mathbf{e}_i^\top + \mathbf{e}_i s_k^\top$ between resolvents. On the spectral event, every resolvent norm along the one-entry segment is bounded by a fixed power of δ^{-1} , so the same holds for each of the finitely many derivatives. For the smooth versions off that event, the spectral-calculus construction in Lemma D.2 gives the same polynomial derivative bound. Hölder's inequality and an arbitrarily large fixed probability exponent bound discrepancies of the exact and smooth derivative identities. Thus the conditional remainder has the asserted form, also when other Green entries or their centered versions occur in f . $\square$

For the next lemma all vector subscripts distinguish factors or coordinates. Fix spectral arguments $z_1, \dots, z_N$ on the deterministic tubes and bounded deterministic top vectors v_j, w_j . Write $G_j = G(z_j)$ and $\widehat{m}_j = \widehat{m}(z_j)$, and define

$$\mathcal{F}_j = v_j^\top (G(z_j) - \mathbb{E}G(z_j)) w_j, \quad m_j = m(z_j), \quad D_j = I_p + m_j \Sigma.$$

For these spectral arguments, define the following two-point quantities:

$$M_{ij} = \Sigma D_i^{-1} D_j^{-1}, \quad Q_{ij} = \frac{m_i m_j}{n} \text{tr}(\Sigma^2 D_i^{-1} D_j^{-1}), \quad S_{ij} = \frac{m_i m_j}{1 - Q_{ij}}.$$

At equal arguments they are defined by continuity. The deterministic leading covariance is

$$\begin{aligned} C_{ij} = & \frac{S_{ij}}{n} \left[(v_i^\top M_{ij} v_j)(w_i^\top M_{ij} w_j) + (v_i^\top M_{ij} w_j)(w_i^\top M_{ij} v_j) \right] \\ & + \frac{\chi_4 m_i m_j}{n} \sum_k (v_i^\top D_i^{-1} s_k)(w_i^\top D_i^{-1} s_k)(v_j^\top D_j^{-1} s_k)(w_j^\top D_j^{-1} s_k). \end{aligned} \quad (\text{E.2})$$

The entries G_{ab}^j denote $a^\top G(z_j) b$. Equation (E.2) is bilinear, without complex conjugation. Its Hermitian covariance form is obtained by conjugating the second spectral argument and its endpoint vectors.

Lemma E.4 (Fixed joint moments). For each fixed $N \geq 2$, uniformly on the deterministic tubes,

$$\mathbb{E} \prod_{j=1}^N \mathcal{F}_j = \sum_{r=2}^N C_{1r} \mathbb{E} \prod_{j \notin \{1, r\}} \mathcal{F}_j + O_N \left(n^{-(N+1)/2 + \epsilon_N} \delta^{-C_N} \right), \quad (\text{E.3})$$

where $\epsilon_N > 0$ can be chosen arbitrarily small by choosing the preliminary exponent sufficiently small. Thus, for even N , we have

$$\mathbb{E} \prod_{j=1}^N \mathcal{F}_j = \sum_{\mathfrak{P}} \prod_{\{i,j\} \in \mathfrak{P}} C_{ij} + O_N(n^{-N/2-1/4} \delta^{-C_N}), \quad (\text{E.4})$$

where $\mathfrak{P}$ ranges over pair partitions. For odd N the whole expectation has the order of the remainder in (E.4). Constants are allowed to depend on fixed N ; no bound for N growing with n is claimed.

Remark E.5. The recursion is the fixed-product version of the method in Yang (2024, Section 4 and Lemma 5.2). That paper derives Gaussian pair partitions for eigenvector resolvent statistics. Here the spectral points are on the explicitly varying contour and the error retains its dependence on δ . The fourth cumulant is kept in the coefficient rather than absorbed into an unspecified covariance. The argument below derives these quantitative statements instead of applying a fixed-parameter limit theorem.

Proof. We apply the cumulant expansion to the first factor, identify terms that pair it with one of the remaining factors, and bound every other derivative allocation by an additional negative power of n . Iterating the resulting recursion proves the pair-partition formula.

Put $t = D_1^{-1} v_1$ and $c_k = t^\top s_k$. Multiply (D.8) by $\mathcal{P} = \prod_{j=2}^N \mathcal{F}_j$, and expand in each independent normalized entry. Centering the first factor instead of subtracting Π costs $O(n^{-1+\epsilon} \delta^{-C}) \mathbb{E} |\mathcal{P}|$, which is $O(n^{-(N+1)/2 + C_N \epsilon} \delta^{-C_N})$ by Lemmas D.2–D.4. Expand through order $2N+10$ in Lemma E.2. Its remainder, even with the crude coefficient sum $Cn\sqrt{p}$, is smaller than the remainder in (E.3).

Second cumulant: one pair is removed. The derivative on $G_{w_1}^1$ cancels the last term of (D.8), except for

$$-\mathbb{E}((\widehat{m}_1 - m_1) G_{\Sigma t, w_1}^1 \mathcal{P}) - \frac{1}{n} \mathbb{E}((G_1 P_2 G_1)_{\Sigma t, w_1} \mathcal{P}).$$

The refined mean, trace fluctuation, and the $N - 1$ centered factors bound these terms by the required error. A derivative on $\mathcal{F}_r$ instead gives, after carrying out the latent-coordinate sum exactly,

$$-\frac{1}{n}\mathbb{E}\left[\left((G_r)_{v_r,\Sigma t}(G_1P_2G_r)_{w_1w_r} + (G_r)_{\Sigma t,w_r}(G_1P_2G_r)_{w_1v_r}\right)\prod_{j\neq\{1,r\}}\mathcal{F}_j\right]. \quad (\text{E.5})$$

There is no free coordinate sum left in this expression. The exact identity

$$G_1P_2G_r = \frac{G_1 - G_r}{z_1 - z_r} \quad (\text{E.6})$$

has the spectral derivative interpretation at coincidence. On a small disk in a tube, the derivative bound in (A.3) applied to the coarse error gives the same moment bound for its derivative, with an additional fixed power of δ^{-1} . For separated arguments, direct division gives that bound. Hence the deterministic equivalent of the top block of (E.6) is $S_{1r}M_{1r}$, with L^s error $n^{-1/2+\epsilon}\delta^{-C_s}$. The identity $S_{1r} = (m_1 - m_r)/(z_1 - z_r)$ follows by subtracting (3.2). Replace also the individual top entry in (E.5) by its deterministic equivalent. The result is the first line of (E.2) times the expectation with factors $1, r$ removed. Every replacement error has one extra coarse fluctuation, giving $n^{-(N+1)/2+C_N\epsilon}\delta^{-C_N}$.

Third cumulant: the row sum is essential. When two derivatives hit at most one external factor, the allocations are $(2, 0), (1, 1), (0, 2)$ between the first Green factor and that factor. For $(2, 0)$ use the explicit second derivative in Lemma D.4. The term $2G_{ii}^1G_{s_k s_k}^1G_{iw_1}^1\mathcal{P}$ is potentially largest. Replace its two diagonal entries by deterministic equivalents, which are independent of i , and sum the last mixed entry as $\sqrt{n}G_{e,w_1}^1$. This gains $\sqrt{n}$ relative to an absolute sum. A diagonal replacement error supplies a second coarse fluctuation, providing the same saving. Other terms in this allocation have a bounded endpoint contraction against c_k or at least three mixed entries.

For $(1, 1)$ the only terms with one mixed entry have, up to permutation of the external endpoints, the form $G_{ii}^1G_{s_k w_1}^1G_{v_r s_k}^rG_{iw_r}^r$. Replace the first three factors by deterministic ones and use the same i -sum. The top coefficients have a bounded absolute sum over k . A replacement error either gives an additional fluctuation while retaining this contraction, or replaces both top endpoints and thus gives two additional small factors before a crude $\sqrt{p}$ sum is needed. For $(0, 2)$ the zero-mixed part of the external second derivative is $2G_{v_r s_k}^rG_{ii}^rG_{s_k w_r}^r$; its product with $G_{iw_1}^1$ has precisely the same structure. All other terms have at least three mixed entries.

There is one further allocation: the two derivatives hit distinct external factors. It has three mixed entries, the original one and one from each external derivative. Its top endpoints contract against c_k . Before multiplication by the $N - 3$ remaining centered factors its size is at most $Cn^{-3/2}n(n^{-1/2+\epsilon})^3\delta^{-C} = Cn^{-2+3\epsilon}\delta^{-C}$. Thus every third-cumulant contribution is bounded by the error in (E.3). The common entry cumulant is independent of i , as required by the row-sum argument. This is where the iid assumption, rather than symmetry, is used.

Fourth cumulant: the non-Gaussian covariance term. The exact first derivative is

$$\partial_{ki}G_{iw_1}^1 = -G_{ii}^1G_{s_k w_1}^1 - G_{is_k}^1G_{iw_1}^1.$$

For a differentiated external factor, we have

$$\partial_{ki}^2G_{v_r w_r}^r = 2G_{v_r s_k}^rG_{ii}^rG_{s_k w_r}^r + 2G_{v_r s_k}^rG_{is_k}^rG_{iw_r}^r + 2G_{v_r i}^rG_{s_k s_k}^rG_{iw_r}^r + 2G_{v_r i}^rG_{s_k i}^rG_{s_k w_r}^r.$$

Only the first summand in each display has no mixed-block entry. Taking these two summands corresponds to one derivative on $G_{iw_1}^1$ and two derivatives on the same external factor. The multiplicity three and the factor two cancel the $3!$ in the cumulant coefficient. Replacing these factors

by their deterministic equivalents and summing over i yields the second line of (E.2). The derivative's minus sign and the three negative top deterministic Green factors give a positive sign. The remaining $N - 2$ centered factors are undifferentiated.

A replacement error adds a coarse fluctuation while preserving the summable top endpoints. The other allocations either leave all $N - 1$ centered factors, contain at least two mixed entries after touching at most two external factors, or touch three external factors and contain at least four mixed entries. Each has at least one additional factor $n^{-1/2+C_N\epsilon}$ relative to the leading order $n^{-N/2}$.

Higher cumulants and a complete power count. For a cumulant of order $j \geq 5$, let s_0 derivatives hit the original mixed entry, let d distinct external factors be differentiated with positive orders $s_1, \dots, s_d$, and let t be the number of mixed entries in a resulting product. The rank-two derivative switches blocks, so

$$s_0 + \sum_{a=1}^d s_a = j - 1, \quad t \geq \mathbf{1}_{\{s_0 \text{ is even}\}} + \#\{a : s_a \text{ is odd}\}.$$

There are $N - 1 - d$ undifferentiated centered factors. If a fixed top endpoint is attached to s_k , its coefficient contracts with c_k by the Cauchy–Schwarz inequality. All other factors may be bounded uniformly, with arbitrarily small powers of n obtained from fixed higher moments and a finite polynomial union. The total power is at most

$$n^{-j/2} n n^{-t/2} n^{-(N-1-d)/2} n^{C_N\epsilon} \delta^{-C_N} = n^{-N/2+(3-j+d-t)/2+C_N\epsilon} \delta^{-C_N}. \quad (\text{E.7})$$

An even positive s_a is at least two; hence the integer $j - d + t$ is at least $(j+1+s_0)/2 + \mathbf{1}_{\{s_0 \text{ is even}\}} \geq 4$. Therefore (E.7) has an extra factor $n^{-1/2}$. If no such top endpoint contraction is available, each differentiated external factor has both fixed top endpoints attached to bottom indices, giving $t \geq 1 + 2d$. The remaining latent sum costs at most $\sqrt{p}$, since $\|c\|_2 \leq C$. Replacing 3 by 4 in (E.7) still gives a smaller order, because $4 - j + d - t \leq 3 - j - d \leq -2$.

These counts also cover the nonleading fourth-cumulant terms after the allocation just extracted is removed. For the third cumulant, the exceptions to the crude count are exactly the one-mixed-entry terms handled by the row-sum calculation. The contractions used throughout are $\sum_k |(s_k^\top v)(s_k^\top w)| \leq L\|v\|_2\|w\|_2$, $\sum_k |c_k| \leq \sqrt{p}\|c\|_2$, and $\sum_i G_{iw} = \sqrt{n}G_{\bar{e},w}$. When endpoint vectors involve the resolvent, the first contraction uses their operator-norm bounds; the remaining coarse fluctuations are bounded uniformly before contracting. Hölder's inequality involves only finitely many moments at fixed N .

This proves (E.3). Iteration removes pairs, with each leading coefficient bounded by $Cn^{-1}\delta^{-C}$. All errors retain the extra factor $n^{-1/2+C'_N\epsilon}$ relative to $n^{-N/2}$. Choose the preliminary exponent so that $C'_N\epsilon < 1/4$. Odd moments terminate at a centered factor and have no leading term. This proves (E.4). $\square$

E.2 A uniform integral bound for the leading covariance

Lemma E.6 (Integrated leading kernel). For the calibrated contour and any deterministic continuous weight ψ , the leading covariance of its integrated signal fluctuation, and the leading covariance of its noise integral, have absolute value at most $C\|\psi\|_{\infty,\Gamma_\delta}^2/n$. This bound is uniform as δ decreases in the logarithmic window. The non-Gaussian fourth-cumulant term obeys the same bound separately in absolute value.

Proof. We express each leading covariance as a sum of products of analytic functions, compare its principal kernel with the Hardy-space kernel, and then calculate its contour integral by Fourier coefficients.

Hardy space and the kernel comparison. Write $\Sigma = \sum_j \lambda_j o_j o_j^\top$ with an orthonormal eigenbasis (o_j) , and set $b_j = \lambda_j/L$. The disk coordinate $w = \sqrt{\gamma}L/(L - z)$ takes the observation contour to $|w| = (1 + \delta)^{-1}$. Define

$$f_j(w) = \frac{1}{\sqrt{p}} \frac{b_j w}{1 - (1 - b_j)w/\sqrt{\gamma}}, \quad K_f(w, \xi) = \frac{1}{1 - \sum_j f_j(w) \overline{f_j(\xi)}}.$$

Since $1 - (1 - b_j)/\sqrt{\gamma} \geq b_j$, we have $\sup_{|w| < 1} \sum_j |f_j(w)/w|^2 \leq 1$.

The Hardy space $H^2(\mathbb{D})$ is the Hilbert space of holomorphic functions $h(w) = \sum_{a \geq 0} h_a w^a$ with $\|h\|_{H^2}^2 = \sum_{a \geq 0} |h_a|^2 < \infty$. Its inner product is linear in the first argument. Evaluation at ξ is represented by $k_\xi(w) = (1 - w\bar{\xi})^{-1}$, as follows by summing the coefficient inner product. For two kernels K_1, K_2 , the notation $K_1 \preceq K_2$ means

$$\sum_{a,b=1}^N c_a \bar{c}_b (K_2(w_a, w_b) - K_1(w_a, w_b)) \geq 0$$

for every finite choice of points and complex coefficients. This is a positive-semidefinite-kernel order, not a pointwise order of complex numbers.

Let $\mathcal{M}_j$ be multiplication by f_j , let S be multiplication by w , and let P_0 project onto constants. For any finite collection $h_j \in H^2$, the pointwise Cauchy–Schwarz inequality on circles, followed by the limit as their radii tend to one, gives $\|\sum_j (f_j/w) h_j\|_{H^2}^2 \leq \sum_j \|h_j\|_{H^2}^2$. It follows that $\sum_j \mathcal{M}_j \mathcal{M}_j^* \preceq SS^* = I - P_0$. For the positive map $\mathcal{T}(A) = \sum_j \mathcal{M}_j A \mathcal{M}_j^*$, telescoping gives

$$\sum_{a=0}^N \mathcal{T}^a(P_0) \preceq \sum_{a=0}^N \mathcal{T}^a(I - \mathcal{T}(I)) = I - \mathcal{T}^{N+1}(I) \preceq I.$$

The kernel of the left side is the partial geometric series in $\sum_j f_j(w) \overline{f_j(\xi)}$. Letting $N \rightarrow \infty$ proves

$$K_f(w, \xi) \preceq \frac{1}{1 - w\bar{\xi}}.$$

If $A(w)$ is analytic and bounded by C , multiplication by A on H^2 has norm at most C , by the same circle argument. Multiplying both kernel arguments by A and $\bar{A}$ therefore preserves this comparison with an additional factor C^2 . These elementary Hardy-space facts have been proved here to fix the conventions used in the integral.

The covariance factors and the contour parameter. After diagonalizing only the deterministic covariance matrices, the signal covariance is a sum of K_f/n multiplied by

$$A_{jk}(z) \overline{A_{jk}(z')}, \quad A_{jk}(z) = \frac{m(\zeta(z)) \sqrt{\lambda_j \lambda_k}}{(1 + \lambda_j m(\zeta(z)))(1 + \lambda_k m(\zeta(z)))}.$$

For an orthonormal eigenbasis $o_1, \dots, o_p$ of Σ , the nonnegative coefficients are

$$((o_j^\top v)(o_k^\top \beta) + (o_k^\top v)(o_j^\top \beta))^2 \quad (j < k), \quad 2((o_j^\top v)(o_j^\top \beta))^2 \quad (j = k).$$

Their sum is $\|v\|_2^2 \|\beta\|_2^2 + (v^\top \beta)^2 \leq 2B_0^2$. For the noise, use $A_j(z) = m(\zeta(z))\sqrt{\lambda_j}/(1 + \lambda_j m(\zeta(z)))$ and coefficients $\sigma^2(o_j^\top v)^2$, which sum to at most $\bar{\sigma}^2$. Since $m(\zeta(z)) = -1/z$, all these factors, as functions of w , are analytic and uniformly bounded on a disk strictly larger than the unit disk.

To integrate, parameterize the contour counterclockwise by $z(t) = L - R_\delta e^{it}$, $0 \leq t \leq 2\pi$. Then $w(t) = (1 + \delta)^{-1} e^{-it}$ and $z'(t) = -iR_\delta e^{it}$. For one analytic multiplier A , define the ordinary periodic function

$$b(t) = \frac{z'(t)}{i} \psi(z(t)) A(z(t)), \quad \hat{b}(j) = \frac{1}{2\pi} \int_0^{2\pi} b(t) e^{-ijt} dt.$$

The factor $z'(t)/i$ is exactly the change from $dz/(2\pi i)$ to $dt/(2\pi)$ in the contour integral. It satisfies $|z'(t)/i| = R_\delta$, so $\|b\|_{L^2(dt/(2\pi))} \leq C\|\psi\|_{\infty, \Gamma_\delta}$. Expanding the geometric series and using Parseval's identity (Whittaker and Watson, 1920, Section 9.5, pp. 180–182) gives

$$\int_0^{2\pi} \int_0^{2\pi} \frac{b(t) \overline{b(s)}}{1 - (1 + \delta)^{-2} e^{-i(t-s)}} \frac{dt ds}{(2\pi)^2} = \sum_{j \geq 0} (1 + \delta)^{-2j} |\hat{b}(j)|^2 \leq \frac{1}{2\pi} \int_0^{2\pi} |b(t)|^2 dt.$$

The equality can first be verified for finite Fourier sums and then obtained by L^2 approximation; it is the coefficient form of Parseval's identity. Summing the bounded nonnegative covariance coefficients proves the required $C\|\psi\|_{\infty, \Gamma_\delta}^2/n$ bound.

Fourth cumulants and products without conjugation. The fourth-cumulant contribution is $\chi_4 \sum_j c_j(z) \overline{c_j(z')}/n$, where

$$c_j(z) = m(\zeta(z)) [\Sigma^{1/2} (I + m(\zeta(z)) \Sigma)^{-1} v]_j [\Sigma^{1/2} (I + m(\zeta(z)) \Sigma)^{-1} \beta]_j.$$

The two vectors in brackets have bounded Euclidean norm on the contour. The Cauchy–Schwarz inequality implies $\sum_j |c_j(z)|^2 \leq C$. Consequently, we have

$$\begin{aligned} & \frac{|\chi_4|}{n} \sum_j \left| \frac{1}{2\pi i} \oint \psi(z) c_j(z) dz \right|^2 \\ & \leq \frac{|\chi_4|}{n} \left(\frac{1}{2\pi} \oint |\psi(z)| \left(\sum_j |c_j(z)|^2 \right)^{1/2} |dz| \right)^2 \leq \frac{C}{n} \|\psi\|_{\infty, \Gamma_\delta}^2. \end{aligned}$$

This uses Minkowski's integral inequality for the Euclidean norm and is valid for either sign of χ_4 .

For clarity, the same bound also covers the bilinear, nonconjugated coefficients in a product of real and imaginary parts. The principal kernel has the convergent feature expansion $K_f(w, \xi) = \sum_{r \geq 0} \sum_{j_1, \dots, j_r} \prod_{a=1}^r f_{j_a}(w) \overline{f_{j_a}(\xi)}$. After incorporating a multiplier and integrating, denote these feature integrals by $c_{r, j_1, \dots, j_r}(\psi)$. The calculation above bounds the sum of their squared absolute values by $C\|\psi\|_{\infty}^2$. For two weights ψ_1, ψ_2 , both $\sum c(\psi_1) \overline{c(\psi_2)}$ and $\sum c(\psi_1) c(\psi_2)$ are bounded in absolute value by $C\|\psi_1\|_{\infty} \|\psi_2\|_{\infty}$, by the Cauchy–Schwarz inequality for these sequences. Real symmetry of the deterministic resolvent identifies the second expression with the covariance without conjugation. The fourth-cumulant term has the identical argument with its finite sequence of integrals. Complex endpoint vectors can be decomposed into real and imaginary parts before applying these bounds. Thus every pair coefficient used in the fixed-moment expansion has the asserted absolute integrated bound. $\square$

E.3 The integrated field and its fixed moments

Proof of Lemma E.1. We integrate the centered design moments, bound the conditional Gaussian noise, and finally include the mean and inverse-calibration errors.

Integrated design fluctuation. First evaluate at deterministic calibrated arguments. Apply Lemma E.4 to $2s$ centered bilinear transfer factors, where $2s \geq r$ is a fixed even integer. The added identity in the top block of (A.1) does not affect centering. Integrate all $2s$ arguments, multiplying by the contour weights. Every pair partition factors into s integrated leading covariances. Lemma E.6 bounds their product by $C_s n^{-s} \|\psi\|_\infty^{2s}$. The integrated remainder is at most $C_s n^{-s-1/4} \delta^{-C_s} \|\psi\|_\infty^{2s}$. Since $\delta^{-C_s} \leq (\log n)^{C_s}$, it is absorbed into the same bound. Real and imaginary parts are obtained by expanding each as half the sum or difference of a field and its conjugate. Lyapunov's inequality then gives the required r th-moment bound for the integrated centered design field.

Conditional noise covariance. At deterministic arguments ξ, η , the bilinear conditional second moment of the noise fields is $\sigma^2 v^\top \mathcal{X}(\hat{S} - \xi I)^{-1}(\hat{S} - \eta I)^{-1} \mathcal{X}^\top v / n$. By (E.6), it is the divided difference of the transfer. The coarse anisotropic estimate in Lemma D.2, transferred to a high-probability uniform bound on the tubes by high moments and a net, replaces it by $\sigma^2 S_{12} v^\top M_{12} v / n$, with error $O(n^{-3/2+\epsilon} \delta^{-C})$. Near coincident arguments use the derivative bound in (A.3) on a smaller disk instead of division by an arbitrarily small difference. The leading deterministic integrated covariance is bounded by Lemma E.6; the error is smaller. Hence, on an X -measurable event,

$$\text{Var} \left(\text{Re} \frac{1}{2\pi i} \oint \psi(z) (\hat{F}(z) - \mathbb{E}(\hat{F}(z) \mid X)) dz \mid X \right) \leq C \sigma^2 \|\psi\|_\infty^2 / n.$$

The same holds for the imaginary part. Conditional Gaussianity converts this variance bound to every fixed absolute moment. It does not require independent coordinate estimates.

Mean and empirical calibration. The refined deterministic-parameter mean error from Lemma D.4 contributes at most $C \|\psi\|_\infty n^{-1+\epsilon} \delta^{-C}$. Lemma D.5 changes the signal part by at most $C \|\psi\|_\infty n^{-3/4} \delta^{-C}$ on the common root event, using the transfer derivative norm on the spectral tube. The coefficient row of the noise change has norm at most $C \|\psi\|_\infty n^{-5/4} \delta^{-C}$, since $\|X/\sqrt{n}\|_{\text{op}} \leq C$. Its conditional Gaussian moments are of smaller order than $n^{-r/2} \|\psi\|_\infty^r$. These errors can all be absorbed because an extra negative power of n dominates any fixed power of $\log n$.

Intersect the spectral, trace, and directional events with failure exponent chosen larger than the desired D and any fixed net or loading-set size. The event depends only on X and so does not truncate the Gaussian noise. On it the smooth deterministic field equals the actual field before root replacement. On its complement polynomial smooth bounds and high moments justify the expectations used in Lemma E.4. The implemented estimator does not inspect this parameter-dependent event; it uses only the branch rule and clipping. Combining the three parts proves (E.1). $\square$

F The scalar lower-bound experiments

We first take $\ell = \kappa^{-1}$, $L = 1$, and $v = e_1$, so the target is β_1 . In the final regression, write $X = (X_1, X_{-1})$ and $\beta = (\beta_1, \beta_{-1}^\top)^\top$, where $X_1 \in \mathbb{R}^n$ and $X_{-1} \in \mathbb{R}^{n \times (p-1)}$. We construct this experiment from a regression with only $p-1$ covariates. The latter design and response are denoted by (X, Y) throughout this Section.

F.1 Definition of the two hypotheses

Set $d = p-1$, $\gamma^\circ = d/n = \gamma - 1/n$, and $v = n^{-1/8}$. The preliminary experiment has the same sample size n and d covariates. Under (2.2), γ° remains uniformly larger than one and bounded above for all sufficiently large n . The final experiment, after adding the constructed first covariate,

has dimension p and ratio γ as required by $\mathcal{G}(n, p, \gamma, \kappa)$. Define

$$\rho = \frac{\sqrt{\gamma^\circ} - 1 + \kappa^{-1} + v}{\sqrt{\gamma^\circ} + 1 - \kappa^{-1} - v}, \quad h = \frac{\pi \mathbf{K}(\rho^2)}{\mathbf{K}(1 - \rho^2)}.$$

The number h is the logarithmic annulus width for the disk $|z - 1| < \sqrt{\gamma^\circ}$ with the interval $[\kappa^{-1} + v, 1]$ removed. To distinguish this geometry from the upper-bound geometry, let $\mathbf{u}$ solve

$$\Delta \mathbf{u} = 0 \quad \text{on } \{z : |z - 1| < \sqrt{\gamma^\circ}\} \setminus [\kappa^{-1} + v, 1], \quad \mathbf{u}|_{[\kappa^{-1} + v, 1]} = 0, \quad \mathbf{u}|_{|z-1|=\sqrt{\gamma^\circ}} = 1.$$

Let $\tilde{\mathbf{u}}$ be its harmonic conjugate with $\tilde{\mathbf{u}}(0) = 0$, and set $\mathbf{q} = h(\mathbf{u} + i\tilde{\mathbf{u}})$. Lemma G.1, applied with the displayed radius and interval, gives imaginary period $2\pi i$ and the real target value

$$\mathbf{q}(0) = \frac{\pi}{\mathbf{K}(1 - \rho^2)} \left[\mathbf{K}(\rho^2) - \mathbf{F} \left(\arcsin \frac{\sqrt{\gamma^\circ} - 1}{\rho(\sqrt{\gamma^\circ} + 1)} \middle| \rho^2 \right) \right].$$

Thus $\mathbf{q}$ is a different, explicitly specified potential; the symbol q_δ continues to refer only to the upper-bound construction.

Spectral weights. Choose

$$K = \left\lceil \frac{\mathbf{K}(1 - \rho^2) \log n}{2\pi \mathbf{K}(\rho^2)} \right\rceil = \left\lceil \frac{\log n}{2h} \right\rceil. \quad (\text{F.1})$$

Take the positive finite atomic measures ν_0, ν_1 supplied by Lemma F.3. List the distinct points in their two supports as

$$\text{supp}(\nu_0) \cup \text{supp}(\nu_1) = \{\lambda_1, \dots, \lambda_s\} \subset (\kappa^{-1} + v, 1).$$

For $j = 0, 1$, let $w_{jk} = \nu_j(\{\lambda_k\})$, so that $\nu_j = \sum_{k=1}^s w_{jk} \delta_{\lambda_k}$. The weights $w_{jk} \geq 0$ are the masses of these measures, including zero when a point belongs only to the other support. The construction gives $s \leq CK$. Define

$$\mathbf{D} = \text{diag} \left(\frac{1 - \lambda_1}{\lambda_1}, \dots, \frac{1 - \lambda_s}{\lambda_s} \right), \quad t_k = \frac{d - s - 1 - n(1 - \lambda_k)}{d}.$$

For all sufficiently large n , we have $\min_k t_k \geq c > 0$, because γ° is bounded away from one and $s = O(\log n)$.

The mixing distribution. On $V \in \mathbb{R}^{d \times s}$, define the probability density

$$\pi(V) = \frac{1}{Z_\pi} \det(I_d + V \mathbf{D} V^\top)^{-n/2} \exp \left(-\frac{d}{2} \text{tr}(\text{diag}(t_1, \dots, t_s) V^\top V) \right). \quad (\text{F.2})$$

The corresponding covariance is $\Sigma_-(V) = (I_d + V \mathbf{D} V^\top)^{-1}$. Here $Z_\pi \in (0, \infty)$ is the normalizing integral. It is finite because the determinant factor is at most one and the remaining factor has a positive definite Gaussian exponent. Write $\mathbf{Q} = V^\top V$, $\mathcal{C} = \{V : \|\mathbf{Q} - I_s\|_{\text{op}} \leq v^2\}$, and $\pi_{\mathcal{C}}(dV) = \pi(dV | \mathcal{C})$. Lemma F.5 shows that $\pi(\mathcal{C}) > 0$ and bounds $\pi(\mathcal{C}^c)$.

Conditional regressions and the final data laws. Set $s_0 = (1 + \kappa^{-1})/4$. For a fixed amplitude $\tau > 0$, to be chosen sufficiently small, define

$$\vartheta_j = (\sqrt{\tau w_{j1}}, \dots, \sqrt{\tau w_{js}})^\top, \quad \varpi_j = s_0 - \tau \int \lambda \nu_j(d\lambda), \quad b_j = \frac{\varpi_j - s_0}{\varpi_j + s_0}, \quad j = 0, 1.$$

Choose τ so that $s_0/2 \leq \varpi_j \leq s_0$. Given V , the $(p-1)$ -covariate regression is

$$\mathbf{X}_i \sim N_d(0, \Sigma_-(V)), \quad \mathbf{Y}_i = \mathbf{X}_i^\top V \vartheta_j + \varepsilon_i^\circ, \quad \varepsilon_i^\circ \sim N(0, \varpi_j),$$

independently for $i = 1, \dots, n$, with the errors independent of $\mathbf{X}$. Its data law mixed over $V \sim \pi$ is denoted by $\tilde{P}_j$, and its data law mixed over $V \sim \pi_{\mathcal{C}}$ by $\tilde{P}_j^{\mathcal{C}}$.

Generate $\xi \sim N(0, s_0 I_n)$ independently, and apply the same transformation under both hypotheses:

$$X_1 = Y + \xi, \quad X_{-1} = X, \quad Y = Y - \xi. \quad (\text{F.3})$$

Given V , the final Gaussian regression has parameters

$$\begin{aligned} \Sigma_j(V) &= \begin{pmatrix} \vartheta_j^\top V^\top \Sigma_-(V) V \vartheta_j + \varpi_j + s_0 & \vartheta_j^\top V^\top \Sigma_-(V) \\ \Sigma_-(V) V \vartheta_j & \Sigma_-(V) \end{pmatrix}, \\ \beta^{(j)}(V) &= (b_j, (1 - b_j)(V \vartheta_j)^\top)^\top, \quad \sigma_j^2 = \frac{4\varpi_j s_0}{\varpi_j + s_0}. \end{aligned} \quad (\text{F.4})$$

Let $\mathbb{Q}_j$ be the distribution of these parameters when $V \sim \pi_{\mathcal{C}}$. Finally define P_j to be the law of the full data under this prior: for every measurable data event E ,

$$P_j(E) = \int \Pr_{\Sigma_j(V), \beta^{(j)}(V), \sigma_j^2}((X, Y) \in E) \pi_{\mathcal{C}}(dV), \quad j = 0, 1.$$

The inner probability is the product Gaussian regression experiment with the fixed displayed parameter.

Remark F.1. Kong and Valiant (2018, Appendix D.2.1, proof of Propositions 14–15 and Corollary 5) use Gaussian regressions with $\Sigma = (I + \sum_k a_k v_k v_k^\top)^{-1}$, a coefficient in the span of the v_k , and a determinant factor $\det(I + \sum_k a_k v_k v_k^\top)^{-n/2}$ in the mixing density. They compare conditional response laws by matching finitely many spectral moments, following Verzelen and Gassiat (2018, proof of Theorem 4.5). The determinant cancellation in (F.2) is the same mechanism. Here the matrices $\mathbf{D}$ and the density π are identical under the two hypotheses. The factors t_k place the mode of $V^\top V$ exactly at I_s . The measures ν_j control a rational transform on the complete spectral support, rather than only finitely many monomials. Their two endpoint-weighted bounds ensure the exact inequalities $\kappa^{-1} I_p \preceq \Sigma_j(V) \preceq I_p$ after (F.3). These are the modifications needed for the stated proportional-dimensional rate.

F.2 The supported priors and their separation

Lemma F.2 (Supported final priors). With the degree (F.1), there is a fixed $\tau > 0$ such that, for all sufficiently large n and every $V \in \mathcal{C}$, the parameters in (F.4) satisfy

$$\begin{aligned} \kappa^{-1} I_p &\preceq \Sigma_j(V) \preceq I_p, \quad \|\beta^{(j)}(V)\|_2 \leq C\sqrt{\tau}, \\ \frac{1 + \kappa^{-1}}{3} &\leq \sigma_j^2 \leq \frac{1 + \kappa^{-1}}{2}, \quad j = 0, 1. \end{aligned}$$

Their data laws satisfy $\text{TV}(P_0, P_1) \leq 1/3$. Moreover,

$$\mathbb{Q}_j(\beta_1 = b_j) = 1, \quad |b_1 - b_0| \geq ce^{-Kq(0)} \asymp n^{-\alpha(\kappa, p/n)/2}.$$

Proof. We verify the regression transformation, bound the two covariance Schur complements, and then combine the conditional comparison with the prior-conditioning error.

The regression parameters. In (F.3), subtracting $b_j X_1 + (1 - b_j) X V \vartheta_j$ from Y gives

$$(1 - b_j) \varepsilon^\circ - (1 + b_j) \xi.$$

Its covariance with X_1 is $((1 - b_j) \varpi_j - (1 + b_j) s_0) I_n = 0$, and its covariance with X is zero. Conditional on V , joint Gaussianity gives independence from the entire design. Its variance is $4 \varpi_j s_0 / (\varpi_j + s_0)$, proving (F.4). On $\mathcal{C}$, the identity

$$V^\top \Sigma_-(V) V = (Q^{-1} + D)^{-1}$$

yields

$$\vartheta_j^\top V^\top \Sigma_-(V) V \vartheta_j = \tau \int \lambda \nu_j(d\lambda) + O(\tau v^2), \quad \|V \vartheta_j\|_2 \leq C \sqrt{\tau}.$$

Thus the upper-left entry of $\Sigma_j(V)$ is $2s_0 + O(\tau v^2)$.

The upper and lower eigenvalue bounds. Write $V = O_V Q^{1/2}$ with $O_V^\top O_V = I_s$, and put $S = (I_s + Q^{1/2} D Q^{1/2})^{-1}$. The matrix $\Sigma_-(V)$ is S on $\text{col}(V)$ and the identity on its orthogonal complement. The off-diagonal block of $\Sigma_j(V)$ lies in $\text{col}(V)$. The upper Schur-complement term is

$$\vartheta_j^\top Q^{1/2} S (I_s - S)^{-1} S Q^{1/2} \vartheta_j \leq \vartheta_j^\top D^{-1} \vartheta_j = \tau \int \frac{\lambda}{1 - \lambda} \nu_j(d\lambda) \leq C \tau.$$

Here $S(I_s - S)^{-1} S = (Q^{1/2} D Q^{1/2})^{-1} - S$, and all retained atoms satisfy $\lambda_k < 1$.

For the lower bound define $C = (1 - \kappa^{-1}) Q^{-1} - \kappa^{-1} D$. Its difference from $\text{diag}((\lambda_k - \kappa^{-1})/\lambda_k)$ has norm $O(v^2)$. Since each $\lambda_k - \kappa^{-1} > v$, we have, for large n ,

$$C \succeq \frac{1}{2} \text{diag} \left(\frac{\lambda_k - \kappa^{-1}}{\lambda_k} \right) \succeq \frac{1}{2} \text{diag} \left(\frac{\lambda_k - \kappa^{-1} - v}{\lambda_k} \right) \succ 0.$$

This also implies $S \succ \kappa^{-1} I_s$. Since $S \preceq I_s$, diagonalizing S proves $S(S - \kappa^{-1} I_s)^{-1} S \preceq S(S - \kappa^{-1} I_s)^{-1}$. Consequently the lower Schur-complement term is at most

$$\vartheta_j^\top C^{-1} \vartheta_j \leq C \tau \int \frac{\lambda}{\lambda - \kappa^{-1} - v} \nu_j(d\lambda) \leq C \tau.$$

The weighted bounds used here are from Lemma F.3. The criterion itself follows by completing the square in a quadratic form: when $D \succ 0$, the block matrix $\begin{pmatrix} a & b^\top \\ b & D \end{pmatrix}$ is positive semidefinite exactly when $a \geq b^\top D^{-1} b$. Apply it on $\text{col}(V)$; the orthogonal identity directions have zero coupling. Since $2s_0 = (1 + \kappa^{-1})/2$, both gaps $2s_0 - \kappa^{-1}$ and $1 - 2s_0$ are fixed positive numbers. Taking τ sufficiently small and then n large proves $\kappa^{-1} I_p \preceq \Sigma_j(V) \preceq I_p$.

The formulas also give $|b_j| \leq C \tau$ and $|1 - b_j| \leq C$, so $\|\beta^{(j)}(V)\|_2 \leq C \sqrt{\tau}$. Since $\varpi_j \in [s_0/2, s_0]$, we have $4s_0/3 \leq \sigma_j^2 \leq 2s_0$, which is the stated interval.

Distance and separation. For any prior π and sampling kernel, conditioning on $\mathcal{C}$ changes the induced data law by at most $\pi(\mathcal{C}^c)$; see Lemma H.4. Thus, we have

$$\text{TV}(\tilde{P}_j^{\mathcal{C}}, \tilde{P}_j) \leq \pi(\mathcal{C}^c).$$

The common random transformation (F.3) does not increase total variation by the data-processing inequality. Lemmas F.6 and F.5 therefore give

$$\text{TV}(P_0, P_1) \leq \text{TV}(\tilde{P}_0, \tilde{P}_1) + 2\pi(\mathcal{C}^c) \leq \frac{1}{3}$$

for all sufficiently large n . The first coefficient is b_j , independently of V , and

$$|b_1 - b_0| = \frac{2s_0\tau \left| \int \lambda(\nu_1 - \nu_0)(d\lambda) \right|}{(2s_0 - \tau \int \lambda d\nu_0)(2s_0 - \tau \int \lambda d\nu_1)} \geq ce^{-Kq(0)}.$$

Finally, the explicit potential gives

$$\frac{q(0)}{h} = \alpha \left((\kappa^{-1} + v)^{-1}, \frac{p-1}{n} \right) = \alpha(\kappa, p/n) + O(v + n^{-1}).$$

The derivative of this formula is bounded on the fixed compact parameter set. Since $(v + n^{-1}) \log n \rightarrow 0$, the degree (F.1) gives $e^{-Kq(0)} \asymp n^{-\alpha(\kappa, p/n)/2}$. $\square$

F.3 Spectral measures with the required transform bound

For $x > 0$, define

$$\mathcal{K}_x(\lambda) = \frac{\lambda^2(\gamma^\circ x - \gamma^\circ + 1 - \lambda)}{\lambda(\gamma^\circ - 1 + \lambda) + \gamma^\circ x(1 - \lambda)}. \quad (\text{F.5})$$

Let $x_- = (1 - (\gamma^\circ)^{-1/2})^2$ and $x_+ = (1 + (\gamma^\circ)^{-1/2})^2$, and define the nonnegative weight

$$\mathcal{W}(x) = \frac{1}{\sqrt{(x - x_-)^2 + K^{-4}}} + \frac{1}{\sqrt{(x - x_+)^2 + K^{-4}}}. \quad (\text{F.6})$$

The interval $[x_-, x_+]$ supports the limiting eigenvalue distribution of $\mathbf{Z}\mathbf{Z}^\top/d$ for an $n \times d$ standard Gaussian matrix $\mathbf{Z}$.

Lemma F.3 (Atomic measures with endpoint control). For $K \asymp \log n$, there are positive finite atomic measures ν_0, ν_1 on $(\kappa^{-1} + v, 1)$ such that

$$\begin{aligned} |\text{supp}(\nu_0) \cup \text{supp}(\nu_1)| &\leq CK, & \nu_0(\mathbb{R}) + \nu_1(\mathbb{R}) &\leq C, \\ \int \left(\frac{\lambda}{1 - \lambda} + \frac{\lambda}{\lambda - \kappa^{-1} - v} \right) \nu_j(d\lambda) &\leq C, & j &= 0, 1, \\ \left| \int \lambda(\nu_1 - \nu_0)(d\lambda) \right| &\geq ce^{-Kq(0)}. \end{aligned}$$

For every $x \in [x_- - v, x_+ + v]$, they also satisfy

$$\left| \int \mathcal{K}_x(\lambda)(\nu_1 - \nu_0)(d\lambda) \right|^2 \leq Ce^{-2Kh}\mathcal{W}(x). \quad (\text{F.7})$$

Remark F.4. The construction is a Cauchy-transform version of the moment-separation step in Kong and Valiant (2018, Appendix D.2.1, Proposition 16). The difference is the endpoint factor $(\lambda - \kappa^{-1} - v)(1 - \lambda)/\lambda^2$ and the weighted bound (F.7). The latter is needed when the two poles of the rational transform coincide at a spectral edge.

Proof. We form a signed measure from the two boundary traces of an analytic function, multiply it by an endpoint factor, and then replace its positive parts by finitely supported measures while preserving the target integral.

Boundary traces and the target integral. Set $f(z) = e^{-Kq(z)}$. Its two slit traces define the real signed measure

$$\eta(d\lambda) = \frac{f_+(\lambda) - f_-(\lambda)}{2\pi i} d\lambda, \quad \mathcal{J}(z) = \int \frac{\eta(d\lambda)}{\lambda - z}.$$

We have $\|\eta\|_{\text{TV}} \leq (1 - \kappa^{-1} - v)/\pi$, because both traces of f have modulus one. Cauchy's integral formula on the slit domain, with the upper side oriented towards 1, gives

$$\mathcal{J}(z) = f(z) - \mathcal{E}(z), \quad \mathcal{E}(z) = \frac{1}{2\pi i} \oint_{|\xi-1|=\sqrt{\gamma^\circ}} \frac{f(\xi)}{\xi - z} d\xi.$$

This is the finitely connected form of the residue theorem; see Ahlfors (1979, Chapter 4, Section 5.1, Theorem 17). Reflection across the outer circle extends $\mathbf{q}$ to a fixed exterior collar. Its outward normal derivative is positive there: the annulus map is conformal at the analytic boundary, and its radial logarithm increases outwards. Continuity of the explicit elliptic formula on the compact parameter family gives a positive uniform lower bound for that derivative. Hence a slightly larger fixed circle $\mathcal{C}_+$ has $\text{Re } \mathbf{q} \geq h + c_0$, with $c_0 > 0$ independent of n .

Deform the contour defining $\mathcal{E}$ to $\mathcal{C}_+$. If a compact neighborhood of the original disk is at distance at least $d_0 > 0$ from $\mathcal{C}_+$, then Cauchy's integral formula and its derivative give the explicit bounds

$$|\mathcal{E}^{(j)}(z)| \leq \frac{j! \text{length}(\mathcal{C}_+)}{2\pi d_0^{j+1}} e^{-K(h+c_0)}, \quad j = 0, 1.$$

The same contour deformation gives $|\int \eta| + |\int \lambda \eta(d\lambda)| \leq C e^{-K(h+c_0)}$.

Multiply η by

$$\frac{(\lambda - \kappa^{-1} - v)(1 - \lambda)}{\lambda^2}.$$

The target integral of the resulting signed measure equals

$$-\int \lambda \eta(d\lambda) + (1 + \kappa^{-1} + v) \int \eta - (\kappa^{-1} + v) \mathcal{J}(0) = -(\kappa^{-1} + v) e^{-K\mathbf{q}(0)} + O(e^{-K(h+c_0)}).$$

Since $0 < \mathbf{q}(0) < h$ uniformly, this proves the required separation. Multiplication of the endpoint factor by $\lambda/(1-\lambda)$ leaves $(\lambda - \kappa^{-1} - v)/\lambda \leq 1$. Multiplication by $\lambda/(\lambda - \kappa^{-1} - v)$ leaves $(1-\lambda)/\lambda \leq C$. Therefore both positive parts have the stated mass and weighted bounds.

The rational transform near the spectral edges. For $x = 1 + (\gamma^\circ)^{-1} + 2(\gamma^\circ)^{-1/2} \cos t$, the denominator of (F.5) factors as

$$(\lambda - z_+)(\lambda - z_-), \quad z_\pm = 1 + \sqrt{\gamma^\circ} e^{\pm it}.$$

Multiplication by the endpoint factor produces a cubic numerator and a quadratic denominator. Its polynomial part is $\lambda - (1 + \kappa^{-1} + v)$; its remainder is

$$(\kappa^{-1} + v - \gamma^\circ x)\lambda + (1 + \kappa^{-1} + v)\gamma^\circ x - (\kappa^{-1} + v)(\gamma^\circ x - \gamma^\circ + 1).$$

Thus the integral is a bounded linear combination of $\int \eta$, $\int \lambda d\eta$, and

$$\frac{\mathcal{J}(z_+) - \mathcal{J}(z_-)}{z_+ - z_-}, \quad \frac{z_+ \mathcal{J}(z_+) - z_- \mathcal{J}(z_-)}{z_+ - z_-}.$$

At coincident roots these expressions are the derivatives of $\mathcal{J}$ and $z\mathcal{J}(z)$.

On the critical circle, $|f(z_\pm)| = e^{-Kh}$. If $|\sin t| \geq K^{-1}$, the first divided difference has absolute value at most $Ce^{-Kh}/|\sin t|$. If $|\sin t| < K^{-1}$, the segment between the two roots is at distance $O(K^{-2})$ from the circle. On that segment, $\text{Re } \mathbf{q} \geq h - CK^{-2}$ and $|\mathbf{q}'| \leq C$, so $|f'| \leq CK e^{-Kh}$. The bounds on $\mathcal{E}$ give the same estimates for $\mathcal{J}$. The second divided difference is bounded by

the first times a bounded root plus a bounded value of $\mathcal{J}$. Hence the transform is bounded by $Ce^{-Kh} \min(K, |\sin t|^{-1})$.

For x at distance at most v beyond an endpoint, the roots are within $C\sqrt{v}$ of their coincident value. The derivative bound is then $CKe^{-Kh+CK\sqrt{v}} \leq C'Ke^{-Kh}$, because $K\sqrt{v} \rightarrow 0$. On the support interval we have $(x - x_-)(x_+ - x) = 4(\gamma^\circ)^{-1} \sin^2 t$. Also $x_+ - x_-$ is bounded above and below, and $v = o(K^{-2})$. These facts imply (F.7) for the continuous signed measure, with the weight (F.6).

Replacement by atomic measures. For each nonzero positive part η^+ or η^- , divide by its mass and integrate the continuous vector

$$(1, \lambda, \dots, \lambda^M, \lambda^{-1}), \quad M = \lceil C_1 K \rceil.$$

The integral lies in the convex hull of the image of the compact interval. Any convex combination with more than $M + 3$ points has affinely dependent vectors. Subtract a suitable multiple of an affine dependence from its nonnegative weights until one weight is zero, and repeat. Limits of such combinations are also in the same compact convex hull. We obtain a positive atomic measure with at most $M + 3$ points and exactly the displayed integrals. This is the finite-dimensional positive-cubature argument underlying Bayer and Teichmann (2006, Theorem 2); the argument just given also allows the additional continuous function λ^{-1} .

Multiply the atomic weights by the endpoint factor. Endpoint atoms then have zero weight and are deleted. The target is unchanged, since its formula uses only the matched integrals of $1, \lambda, \lambda^{-1}$. The weighted bounds remain valid by the pointwise inequalities in Step 1.

The rational integrands in Step 2 are holomorphic and uniformly bounded in a fixed complex neighborhood of $[\kappa^{-1} + v, 1]$: their poles lie near the critical circle, at positive distance from that interval. Lemma H.1 supplies polynomial approximations of degree M with error $C\rho_0^{-M}$ for some fixed $\rho_0 > 1$. Choose C_1 so that this error is at most $e^{-K(h+c_0)}$. Moment matching then changes the transform by at most a constant times this quantity. Since $\mathcal{W}$ has a positive lower bound on the enlarged interval, it is absorbed into (F.7). This gives the asserted atomic ν_0, ν_1 . $\square$

F.4 Concentration of the mixing matrix

Lemma F.5 (Concentration of the full Gram matrix). For the density (F.2), we have

$$\pi(C^c) \leq \exp(-c\sqrt{n} + C(\log n)^3) = o(n^{-D_0}) \quad (\text{F.8})$$

for every fixed $D_0 > 0$. On $\mathcal{C}$ we have $\|V\|_{\text{op}} \leq 2$ and $\lambda_{\min}(\Sigma_-(V)) \geq \kappa^{-1} + v - Cv^2$.

Proof. We compute the density of $\mathbf{Q} = V^\top V$, prove a quantitative decrease from its mode, and compare its tail integral with an explicit lower bound for the normalizing integral.

Density and curvature. Thin QR coordinates give the density of $\mathbf{Q}$ with respect to $\prod_{a \leq b} dQ_{ab}$, up to a normalizing constant, as

$$(\det \mathbf{Q})^{(d-s-1)/2} \det(I_s + \mathbf{D}^{1/2} \mathbf{Q} \mathbf{D}^{1/2})^{-n/2} e^{-d \operatorname{tr}(\operatorname{diag}(t_k) \mathbf{Q})/2}.$$

Indeed, for the upper-triangular QR factor T , the factors in dV and $d\mathbf{Q}$ are proportional to $\prod_k T_{kk}^{d-k}$ and $2^s \prod_k T_{kk}^{s+1-k}$, respectively. Their quotient gives the displayed determinant power. Let ϕ be the logarithm of this density. The choice of t_k gives $\nabla \phi(I_s) = 0$. For a symmetric direction E , put $\mathbf{B} = \mathbf{D}^{1/2}(I_s + \mathbf{D}^{1/2} \mathbf{Q} \mathbf{D}^{1/2})^{-1} \mathbf{D}^{1/2}$. Then $0 \preceq \mathbf{B} \preceq \mathbf{Q}^{-1}$, and

$$\begin{aligned} D^2 \phi(\mathbf{Q})[E, E] &= -\frac{d-s-1}{2} \operatorname{tr}(\mathbf{Q}^{-1} E \mathbf{Q}^{-1} E) + \frac{n}{2} \operatorname{tr}(\mathbf{B} E \mathbf{B} E) \\ &\leq -\frac{d-s-1-n}{2} \operatorname{tr}(\mathbf{Q}^{-1} E \mathbf{Q}^{-1} E). \end{aligned}$$

The last inequality follows from $\mathbf{B} \otimes \mathbf{B} \preceq \mathbf{Q}^{-1} \otimes \mathbf{Q}^{-1}$ and the vectorization formula for the quadratic form; no sign condition on E is used. Since $d - s - 1 - n \geq cd$, integrating along the segment from I_s to $\mathbf{Q}$ gives

$$\phi(\mathbf{Q}) - \phi(I_s) \leq -cd \operatorname{tr}(\mathbf{Q} - I_s - \log \mathbf{Q}). \quad (\text{F.9})$$

Normalizer and tail. On $\|\mathbf{Q} - I_s\|_F \leq d^{-1/2}$, the absolute Hessian is at most Cd , so the integral of the density is at least $e^{\phi(I_s) - C}$ times the volume of this ball in the $s(s+1)/2$ independent matrix entries. Its logarithm is at least $\phi(I_s) - Cs^2 \log d$. If $0 < t < 1/2$ and $\|\mathbf{Q} - I_s\|_{\text{op}} > t$, at least one eigenvalue yields $\operatorname{tr}(\mathbf{Q} - I_s - \log \mathbf{Q}) \geq ct^2$. Split half of the exponential in (F.9) to extract e^{-cdt^2} . Lemma H.5 bounds the integral of the remaining exponential by $e^{Cs^2 \log d}$. Thus, we have

$$\pi(\|\mathbf{Q} - I_s\|_{\text{op}} > t) \leq e^{-cdt^2 + Cs^2 \log d}.$$

Taking $t = v^2$ proves (F.8). Finally, the nonzero eigenvalues of $\mathbf{V}\mathbf{D}\mathbf{V}^\top$ equal those of $\mathbf{D}^{1/2}\mathbf{Q}\mathbf{D}^{1/2}$. The latter differ from the entries of $\mathbf{D}$ by $O(v^2)$. The function $x \mapsto (1+x)^{-1}$ is Lipschitz on the relevant positive interval, which gives the eigenvalue assertion for $\Sigma_-(V)$. $\square$

F.5 Conditional likelihood and total variation

Lemma F.6 (Conditional Gaussian comparison). For the unconditioned data laws $\tilde{P}_0, \tilde{P}_1$, the design marginals coincide. Given $\mathbf{X}$, the columns V_k are independent with

$$V_k \mid \mathbf{X} \sim N_d\left(0, (dt_k I_d + \mathbf{D}_{kk} \mathbf{X}^\top \mathbf{X})^{-1}\right). \quad (\text{F.10})$$

Under hypothesis j , the conditional law of $\mathbf{Y}$ is $N(0, \mathcal{V}_j(\mathbf{X}))$, where

$$\mathcal{V}_j(\mathbf{X}) = \varpi_j I_n + \tau \sum_{k=1}^s w_{jk} \mathbf{X} (dt_k I_d + \mathbf{D}_{kk} \mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

Moreover,

$$\operatorname{TV}(\tilde{P}_0, \tilde{P}_1) \leq C\tau(ne^{-2Kh} + K^2/n)^{1/2} + Ce^{-cnv^2}. \quad (\text{F.11})$$

With (F.1) and sufficiently small fixed τ , this distance is at most $1/4$ for all sufficiently large n .

Proof. The determinant in the Gaussian design likelihood cancels the determinant factor in (F.2); we use the resulting posterior to compute the conditional variances, and then bound their divergence using an integrable spectral-edge weight.

The dependence of the likelihood on V is

$$\det(I_d + \mathbf{V}\mathbf{D}\mathbf{V}^\top)^{n/2} \exp\left(-\frac{1}{2} \operatorname{tr}(\mathbf{X}\mathbf{X}^\top) - \frac{1}{2} \sum_k \mathbf{D}_{kk} \|\mathbf{X}V_k\|_2^2\right).$$

Multiplication by (F.2) proves (F.10). Integrating the linear Gaussian vector $\mathbf{X}V\vartheta_j$ gives $\mathcal{V}_j(\mathbf{X})$. If x is an eigenvalue of $\mathbf{X}\mathbf{X}^\top/d$, the corresponding conditional variance is

$$v_j(x) = \varpi_j + \tau \sum_{k=1}^s \frac{w_{jk}x}{t_k + \mathbf{D}_{kk}x}.$$

The difference is

$$v_1(x) - v_0(x) = \tau \int \left(\frac{x}{1 - (1-\lambda)/\gamma^\circ - (s+1)/d + (1-\lambda)x/\lambda} - \lambda \right) (\nu_1 - \nu_0)(d\lambda).$$

All denominators are bounded below on the fixed positive spectral interval used below. Removing $(s+1)/d$ changes the expression by $O(\tau K/n)$, and the remaining integrand is exactly (F.5).

The sample eigenvalues. Given V , write $\mathbf{X} = \mathbf{Z}\Sigma_-(V)^{1/2}$, with $\mathbf{Z}$ standard Gaussian. Let P_V project onto a $(d-s)$ -dimensional subspace of $\ker V^\top$. Then

$$P_V \preceq \Sigma_-(V) \preceq I_d, \quad \mathbf{Z}P_V\mathbf{Z}^\top/d \preceq \mathbf{X}\mathbf{X}^\top/d \preceq \mathbf{Z}\mathbf{Z}^\top/d.$$

For an $n \times D$ standard Gaussian matrix with $D \geq n$, the singular-value inequality stated in [Zwiernik et al. \(2017, Theorem 3\)](#) gives

$$\Pr \left(s_{\min} < \sqrt{D} - \sqrt{n} - t \text{ or } s_{\max} > \sqrt{D} + \sqrt{n} + t \right) \leq 2e^{-t^2/2}.$$

Apply it with $D = d-s, d$ and $t = c_0 v \sqrt{d}$. Both column counts exceed n and both lower bounds are positive for sufficiently small fixed c_0 and large n . Since $s/d = o(v)$, every eigenvalue x_i of $\mathbf{X}\mathbf{X}^\top/d$ belongs to $[x_- - v, x_+ + v]$ on an event $\mathcal{E}$ satisfying $\tilde{P}_j(\mathcal{E}^c) \leq Ce^{-cnv^2}$.

The weighted spectral average. Lemma F.7 gives $\mathbb{E}[n^{-1} \text{tr} \mathcal{W}(\mathbf{Z}\mathbf{Z}^\top/d)] \leq C$. The two matrices differ by rank at most s . The rank inequality for empirical spectral distribution functions, followed by integration by parts, gives

$$\left| \frac{1}{n} \text{tr} \mathcal{W}(\mathbf{X}\mathbf{X}^\top/d) - \frac{1}{n} \text{tr} \mathcal{W}(\mathbf{Z}\mathbf{Z}^\top/d) \right| \leq \frac{s}{n} \text{Var}_{\mathbb{R}}(\mathcal{W}) \leq \frac{4sK^2}{n} = o(1).$$

The rank inequality follows from eigenvalue interlacing for a rank-one perturbation, iterated s times. Therefore the common design law satisfies $\mathbb{E}[n^{-1} \sum_i \mathcal{W}(x_i)] \leq C$.

Conditional divergence. The matrices $\mathcal{V}_j(\mathbf{X})$ commute. Their eigenvalues $v_j(x_i)$ are bounded above and below on $\mathcal{E}$, and are at least $s_0/2$ everywhere. The centered Gaussian likelihood formula gives

$$D_{\text{KL}}(\tilde{P}_0^{\mathbf{Y}|\mathbf{X}} \| \tilde{P}_1^{\mathbf{Y}|\mathbf{X}}) = \frac{1}{2} \sum_{i=1}^n \left(\frac{v_0(x_i)}{v_1(x_i)} - 1 - \log \frac{v_0(x_i)}{v_1(x_i)} \right).$$

Taylor's theorem on the compact interval of variance ratios, Lemma F.3, and the finite correction above imply

$$\mathbb{E} \left[\mathbf{1}_{\mathcal{E}} D_{\text{KL}}(\tilde{P}_0^{\mathbf{Y}|\mathbf{X}} \| \tilde{P}_1^{\mathbf{Y}|\mathbf{X}}) \right] \leq C\tau^2(ne^{-2Kh} + K^2/n).$$

Because the design marginals agree, total variation of the joint laws is the average of the conditional total variations. Pinsker's inequality on $\mathcal{E}$, the Cauchy–Schwarz inequality for its expectation, and the bound one on $\mathcal{E}^c$ prove (F.11). At the degree (F.1), $ne^{-2Kh} \leq 1$. A sufficiently small fixed τ proves the last assertion. $\square$

Lemma F.7 (Weighted Gaussian covariance trace). Let $\mathbf{Z}$ be $n \times d$ standard Gaussian, with d/n in a compact subset of $(1, \infty)$. For the function (F.6) and $K = O(\log n)$, we have $\mathbb{E}[n^{-1} \text{tr} \mathcal{W}(\mathbf{Z}\mathbf{Z}^\top/d)] \leq C$.

Proof. The deterministic Marchenko–Pastur density is $\gamma^\circ \sqrt{(x-x_-)(x_+-x)}/(2\pi x)$ on (x_-, x_+) . Its lower endpoint is bounded away from zero. Near either endpoint, the product of the density with the corresponding term of $\mathcal{W}$ is bounded by $Ct^{1/2}/\sqrt{t^2 + K^{-4}} \leq Ct^{-1/2}$. Its integral is therefore bounded uniformly in K .

Apply [Bandeira et al. \(2023, Corollary 2.9\)](#) to the Hermitian dilation of $\mathbf{Z}^\top/\sqrt{d}$ and an even smooth compactly supported function agreeing with $t \mapsto \mathcal{W}(t^2)$ on a fixed interval containing

the deterministic spectrum. Its $W^{6,1}$ norm is at most CK^C by differentiation of (F.6). The comparison parameter is $O(n^{-1})$, so the normalized trace error is $O(n^{-1}K^C) = o(1)$. Subtract the $d - n$ structural zero eigenvalues of the dilation to recover the bottom-block trace, and divide by the bounded ratio of dimensions. Outside the cutoff interval, $\mathcal{W} \leq 2K^2$ and the Gaussian singular-value inequality makes the expectation exponentially small. This proves the claim. The result used here is a scalar trace comparison; its exact $W^{6,1}$ form also follows from Lemma H.2. $\square$

G Results on the elliptic integral

This appendix proves the formulas for u_δ and q_δ used in Section 3.1, the geometry of the level curve Λ_δ , and the contour formula (3.22). The identification of a Dirichlet solution with harmonic measure is standard; see Ransford (1995, Sections 4.1 and 4.3). The elliptic-function identities used below are standard formulas for the Jacobi functions; see Whittaker and Watson (1920, Sections 22.11–22.12, 22.2, 22.21, 22.3–22.35, and 22.4).

Lemma G.1 (Potential, period, and level curve). Recall

$$k = \frac{R_\delta - L + \ell}{R_\delta + L - \ell}, \quad T_\delta(z) = \frac{R_\delta - L + z}{R_\delta + L - z},$$

and let b_δ be defined by (3.9). Then $0 < b_\delta(z) < 1$ for $z \in \mathbb{D}_\delta \setminus [\ell, L]$, and b_δ extends continuously with value one on the slit and zero on Γ_δ . Consequently, the function in (3.10) is the unique bounded solution of (3.7).

For the local harmonic conjugate in (3.11), the corresponding local determinations of q_δ agree on overlaps up to integer multiples of $2\pi i$. A counterclockwise circuit once around $[\ell, L]$ increases q_δ by $2\pi i$. Moreover,

$$\operatorname{Re} q_\delta(z) = \frac{\pi \mathbf{K}(k^2)}{\mathbf{K}(1-k^2)} u_\delta(z),$$

so that $\operatorname{Re} q_\delta = 0$ on the slit and $\operatorname{Re} q_\delta = \frac{\pi \mathbf{K}(k^2)}{\mathbf{K}(1-k^2)}$ on Γ_δ . The two traces on the open slit satisfy

$$q_{\delta,-}(x) = -q_{\delta,+}(x) \pmod{2\pi i}, \quad \ell < x < L.$$

At zero, we have

$$q_\delta(0) = \frac{\pi}{\mathbf{K}(1-k^2)} \left[\mathbf{K}(k^2) - \mathbf{F} \left(\arcsin \frac{R_\delta - L}{k(R_\delta + L)} \middle| k^2 \right) \right] > 0. \quad (\text{G.1})$$

For all sufficiently small δ , the level set

$$\Lambda_\delta = \{z \in \mathbb{D}_\delta \setminus [\ell, L] : u_\delta(z) = \delta\}$$

is an analytic Jordan curve whose bounded complementary component contains $[\ell, L]$ but not zero, and the Hausdorff distance

$$\operatorname{dist}_H(\Lambda_\delta, [\ell, L]) \leq C\delta.$$

On the region $\{z : \delta < u_\delta(z) < 1\}$, $\frac{u_\delta(z) - \delta}{1 - \delta}$ is the harmonic measure of Γ_δ . In particular, (3.8) holds, and the formulas extend continuously to $\delta = 0$ with $u_0(0) = \alpha(\kappa, \gamma)$. Under (2.2), after fixing the lower and upper bounds for γ , there are constants $0 < c < C < \infty$ such that, for all sufficiently small δ , $c \leq \alpha_\delta \leq 1 - c$, $c \leq q_\delta(0) \leq C$, $c \leq \frac{\pi \mathbf{K}(k^2)}{\mathbf{K}(1-k^2)} - q_\delta(0) \leq C$. Finally, the contour expression (3.22) follows from these formulas and Jacobi's imaginary transformation.

Remark G.2. The formula (3.10) gives the harmonic measure of the outer boundary of the present slit disk in terms of the incomplete elliptic integral of the first kind. Harmonic measure as the solution of the Dirichlet problem is described in Ransford (1995, Section 4.3). The inverse elliptic integral, addition formulas, periods, and imaginary transformation used to obtain this specialization are the classical Jacobi formulas in Whittaker and Watson (1920, Sections 22.11–22.12, 22.2, 22.21, 22.3–22.35, and 22.4). The formulas for b_δ , u_δ , and q_δ below are derived from those identities for the interval $[\ell, L]$ and circle Γ_δ .

G.1 Optimal exponent asymptotics

The following proposition shows the asymptotic behavior of the exponent $\alpha(\kappa, \gamma)$ when both κ and γ diverge.

Proposition G.3 (Monotonicity and joint asymptotic expansion). The function $\alpha(\kappa, \gamma)$ is strictly decreasing in κ for every fixed $\gamma > 1$, and strictly decreasing in γ for every fixed $\kappa > 1$.

There exist constants $C > 0$ and $\kappa_0, \gamma_0 > 1$ such that, for every $\kappa \geq \kappa_0$ and $\gamma \geq \gamma_0$,

$$\begin{aligned} \frac{\sqrt{\kappa} \log(16\gamma)}{4} \alpha(\kappa, \gamma) &= 1 + \frac{1}{\kappa} \left(\frac{1}{3} - \frac{2}{\log(16\gamma)} \right) \\ &\quad + \frac{1}{\gamma} \left(\frac{1}{4} + \frac{1}{2\log(16\gamma)} \right) + R(\kappa, \gamma), \end{aligned} \quad (\text{G.2})$$

where

$$|R(\kappa, \gamma)| \leq C \left(\kappa^{-2} + \frac{1}{\kappa\gamma} + \gamma^{-2} \right).$$

In particular, after increasing C if necessary,

$$\left| \frac{\sqrt{\kappa} \log(16\gamma)}{4} \alpha(\kappa, \gamma) - 1 \right| \leq C(\kappa^{-1} + \gamma^{-1}) \quad (\text{G.3})$$

throughout this range.

Proof. We prove monotonicity by comparing harmonic measures, and obtain the expansion from an exact integral representation whose numerator remains regular as both parameters diverge.

Monotonicity. For $\kappa, \gamma > 1$, let $u_{\kappa, \gamma}$ be the bounded harmonic function on

$$\mathcal{D}_{\kappa, \gamma} = \{z \in \mathbb{C} : |z - 1| < \sqrt{\gamma}\} \setminus [\kappa^{-1}, 1]$$

with boundary values zero on the slit and one on the circle. Lemma G.1, with $L = 1$, $\ell = \kappa^{-1}$ and $\delta = 0$, gives $\alpha(\kappa, \gamma) = u_{\kappa, \gamma}(0)$. The same lemma gives continuity up to the boundary, including the slit endpoints. We use the maximum principle and its strict interior form; see Ransford (1995, Section 2.3). The interpretation as harmonic measure is the Dirichlet-problem characterization in Ransford (1995, Section 4.3).

Fix $\gamma > 1$ and take $1 < \kappa_1 < \kappa_2$. Then $\mathcal{D}_{\kappa_2, \gamma} \subset \mathcal{D}_{\kappa_1, \gamma}$. The difference $u_{\kappa_1, \gamma} - u_{\kappa_2, \gamma}$ is harmonic on $\mathcal{D}_{\kappa_2, \gamma}$ and has nonnegative boundary values. It vanishes on the outer circle and the common slit. On the added interval $(\kappa_2^{-1}, \kappa_1^{-1})$, however, $u_{\kappa_1, \gamma} > 0$ and $u_{\kappa_2, \gamma} = 0$. Thus the difference is not identically zero, and the strict maximum principle gives $u_{\kappa_1, \gamma}(0) > u_{\kappa_2, \gamma}(0)$.

Next fix $\kappa > 1$ and take $1 < \gamma_1 < \gamma_2$. On $\mathcal{D}_{\kappa, \gamma_1}$, the difference $u_{\kappa, \gamma_1} - u_{\kappa, \gamma_2}$ is harmonic. Its boundary value is zero on the slit and strictly positive on the smaller circle: there $u_{\kappa, \gamma_1} = 1$, while $u_{\kappa, \gamma_2} < 1$ because that circle lies inside the larger domain. The strict maximum principle therefore gives $u_{\kappa, \gamma_1}(0) > u_{\kappa, \gamma_2}(0)$. This proves both monotonicity assertions.

An exact integral representation. Put $a = \kappa^{-1}$ and $b = \gamma^{-1}$. Then

$$\rho = \frac{1 - (1-a)\sqrt{b}}{1 + (1-a)\sqrt{b}}, \quad \xi = \frac{1 - \sqrt{b}}{1 + \sqrt{b}}, \quad 0 < \xi < \rho < 1.$$

The defining elliptic integrals give

$$\mathbf{K}(\rho^2) - \mathbf{F}(\arcsin(\xi/\rho) \mid \rho^2) = \int_{\xi}^{\rho} \frac{dx}{\sqrt{(\rho^2 - x^2)(1 - x^2)}}.$$

Substitute $x = (1 - \sqrt{b}v)/(1 + \sqrt{b}v)$. The integral becomes

$$\frac{1 + (1-a)\sqrt{b}}{2} \int_{1-a}^1 \frac{dv}{\sqrt{v(v-1+a)(1-(1-a)bv)}}.$$

Landen's transformation, in the parameter convention for $\mathbf{K}$, states that

$$\mathbf{K}(\rho^2) = \frac{1 + (1-a)\sqrt{b}}{2} \mathbf{K}(1 - (1-a)^2b);$$

see [Abramowitz and Stegun \(1964, formula 17.3.29\)](#). All arguments belong to $(0, 1)$, so only real positive determinations are involved.

Dividing the preceding two expressions and substituting $v = (1-a)/(1-at^2)$ yields the exact identity

$$\alpha(\kappa, \gamma) = \frac{2\sqrt{a}}{\mathbf{K}(1 - (1-a)^2b)} \int_0^1 \frac{dt}{\sqrt{(1-at^2)(1-(1-a)^2b-at^2)}}. \quad (\text{G.4})$$

This identity holds for every $0 < a, b < 1$.

Uniform expansion of the numerator. For $0 \leq a, b \leq 1/4$ and $0 \leq t \leq 1$, $1 - at^2 \geq \frac{3}{4}$, $1 - (1-a)^2b - at^2 \geq \frac{1}{2}$. Consequently, the integrand in (G.4) and its first two derivatives with respect to (a, b) are uniformly bounded on this set. At $(a, b) = (0, 0)$, its value is one and its first derivatives are t^2 and $1/2$, respectively. Taylor's theorem, followed by integration over t , therefore gives

$$\int_0^1 \frac{dt}{\sqrt{(1-at^2)(1-(1-a)^2b-at^2)}} = 1 + \frac{a}{3} + \frac{b}{2} + O(a^2 + ab + b^2), \quad (\text{G.5})$$

with a constant independent of a, b .

Uniform expansion of the complete elliptic integral. The hypergeometric representation of $\mathbf{K}$ and the convergent continuation formula in [Abramowitz and Stegun \(1964, formulas 17.3.9 and 15.3.10\)](#) give, for $0 < \varepsilon < 1$, we have

$$\mathbf{K}(1 - \varepsilon) = \sum_{j=0}^{\infty} \left(\frac{\binom{2j}{j}}{4^j} \right)^2 \varepsilon^j \left[\log \frac{4}{\sqrt{\varepsilon}} - 2(H_{2j} - H_j) \right],$$

where $H_0 = 0$ and $H_j = \sum_{i=1}^j i^{-1}$. The terms with $j = 0, 1$ are

$$\log \frac{4}{\sqrt{\varepsilon}} \quad \text{and} \quad \frac{\varepsilon}{4} \left(\log \frac{4}{\sqrt{\varepsilon}} - 1 \right).$$

For $j \geq 1$, we have

$$0 \leq H_{2j} - H_j \leq \log 2, \quad 0 < \left(\frac{\binom{2j}{j}}{4^j} \right)^2 \leq 1.$$

Thus, for $0 < \varepsilon \leq 1/4$, the absolute sum of the terms with $j \geq 2$ is bounded by

$$\left(\log \frac{4}{\sqrt{\varepsilon}} + 2 \log 2 \right) \frac{\varepsilon^2}{1 - \varepsilon} \leq C \varepsilon^2 \log \frac{4}{\sqrt{\varepsilon}}.$$

We have therefore established the uniform expansion

$$\mathbf{K}(1 - \varepsilon) = \log \frac{4}{\sqrt{\varepsilon}} + \frac{\varepsilon}{4} \left(\log \frac{4}{\sqrt{\varepsilon}} - 1 \right) + O \left(\varepsilon^2 \log \frac{4}{\sqrt{\varepsilon}} \right). \quad (\text{G.6})$$

Now take $\varepsilon = (1 - a)^2 b$ and write $\mathcal{L} = \log(4/\sqrt{b})$. Uniformly for $0 < a, b \leq 1/4$,

$$\log \frac{4}{\sqrt{\varepsilon}} = \mathcal{L} - \log(1 - a) = \mathcal{L} + a + O(a^2).$$

Since $\varepsilon = b + O(ab)$, substituting into (G.6) gives

$$\mathbf{K}(1 - (1 - a)^2 b) = \mathcal{L} + a + \frac{b}{4} (\mathcal{L} - 1) + O(\mathcal{L}(a^2 + ab + b^2)).$$

In particular, we have

$$\frac{\mathbf{K}(1 - (1 - a)^2 b)}{\mathcal{L}} = 1 + \frac{a}{\mathcal{L}} + \frac{b}{4} \left(1 - \frac{1}{\mathcal{L}} \right) + O(a^2 + ab + b^2). \quad (\text{G.7})$$

Here $\mathcal{L} \geq \log 8$, so the constants remain uniform even when a and b approach zero at different rates.

Combining the expansions. Equations (G.4), (G.5), and (G.7), together with $(1 + x)^{-1} = 1 - x + O(x^2)$ for sufficiently small $|x|$, imply

$$\frac{\mathcal{L}}{2\sqrt{a}} \alpha(\kappa, \gamma) = 1 + a \left(\frac{1}{3} - \frac{1}{\mathcal{L}} \right) + b \left(\frac{1}{4} + \frac{1}{4\mathcal{L}} \right) + O(a^2 + ab + b^2).$$

All remainders are bounded by a single constant throughout a sufficiently small rectangle $0 < a \leq a_0$, $0 < b \leq b_0$. Using $a = \kappa^{-1}$, $b = \gamma^{-1}$, and $2\mathcal{L} = \log(16\gamma)$ proves (G.2) with the stated bound on $R(\kappa, \gamma)$. The coefficients of κ^{-1} and γ^{-1} are uniformly bounded, so (G.3) follows. Finally, $\log(16\gamma)/\log \gamma \rightarrow 1$, which proves (3.19). $\square$

G.2 Jacobi formulas

The Jacobi functions are determined near zero by $s = \mathbf{F}(\phi \mid m)$, $\text{sn}(s \mid m) = \sin \phi$, $\text{cn}(s \mid m) = \cos \phi$, and $\text{dn}(s \mid m) = (1 - m \sin^2 \phi)^{1/2}$, and elsewhere by analytic continuation. We use the parameter m , not the modulus, as their second argument.

Lemma G.4 (Jacobi elliptic-function identities; Whittaker and Watson, 1920, Sections 22.11–22.12, 22.2, 22.21, 22.3–22.35, and 22.4). For $0 < a < 1$, let

$$s_u = \text{sn}(u \mid a), \quad c_u = \text{cn}(u \mid a), \quad d_u = \text{dn}(u \mid a),$$

and define s_v, c_v, d_v analogously. Then we have $s_u^2 + c_u^2 = 1, d_u^2 + a s_u^2 = 1$,

$$\begin{aligned} \text{sn}(\mathbf{F}(\phi \mid a) \mid a) &= \sin \phi, \\ \text{sn}(u + v \mid a) &= \frac{s_u c_v d_v + s_v c_u d_u}{1 - a s_u^2 s_v^2}, \quad \text{cn}(u + v \mid a) = \frac{c_u c_v - s_u s_v d_u d_v}{1 - a s_u^2 s_v^2}, \\ \text{dn}(u + v \mid a) &= \frac{d_u d_v - a s_u s_v c_u c_v}{1 - a s_u^2 s_v^2}. \end{aligned} \quad (\text{G.8})$$

Jacobi's imaginary transformation is

$$\begin{aligned} \operatorname{sn}(iy \mid a) &= i \frac{\operatorname{sn}(y \mid 1-a)}{\operatorname{cn}(y \mid 1-a)}, \quad \operatorname{cn}(iy \mid a) = \frac{1}{\operatorname{cn}(y \mid 1-a)}, \\ \operatorname{dn}(iy \mid a) &= \frac{\operatorname{dn}(y \mid 1-a)}{\operatorname{cn}(y \mid 1-a)}. \end{aligned} \tag{G.9}$$

The function $\operatorname{sn}(\cdot \mid a)$ has periods $4\mathbf{K}(a)$ and $2i\mathbf{K}(1-a)$; hence $\operatorname{sn}^2(\cdot \mid a)$ has periods $2\mathbf{K}(a)$ and $2i\mathbf{K}(1-a)$.

G.3 Proof of the potential formula and level geometry

Proof of Lemma G.1. We first construct a conformal coordinate, then derive the real formula (3.10), and finally verify the period, the level curve, the exponent, and the contour expression (3.22).

The conformal coordinate and harmonic measure. The map

$$T_\delta(z) = \frac{R_\delta - L + z}{R_\delta + L - z}$$

sends $\mathbb{D}_\delta$ to the right half-plane and sends the slit $[\ell, L]$ to $[k, 1]$. Hence $T_\delta(z)^2$ sends $\mathbb{D}_\delta \setminus [\ell, L]$ to

$$\mathbb{C} \setminus ((-\infty, 0] \cup [k^2, 1]).$$

The standard inversion formulas for the Jacobi function give a local holomorphic coordinate U_δ characterized by

$$T_\delta(z) = k \operatorname{sn}(U_\delta(z) \mid k^2), \quad 0 < \operatorname{Re} U_\delta(z) < \mathbf{K}(k^2), \tag{G.10}$$

where different determinations differ by integer multiples of $2i\mathbf{K}(1-k^2)$. One way to fix the normalization is to start on the real interval $L - R_\delta < z < \ell$, where $0 < T_\delta(z) < k$, and set

$$U_\delta(z) = \mathbf{F} \left(\arcsin \frac{T_\delta(z)}{k} \mid k^2 \right). \tag{G.11}$$

The inverse and period statements in (G.10) are the Jacobi inversion and period formulas cited in Lemma G.4.

The outer circle corresponds to $\operatorname{Re} U_\delta = 0$, while the two sides of the slit correspond to $\operatorname{Re} U_\delta = \mathbf{K}(k^2)$. Therefore

$$1 - \frac{\operatorname{Re} U_\delta(z)}{\mathbf{K}(k^2)} \tag{G.12}$$

is single-valued and harmonic on the slit disk, extends continuously with value one on Γ_δ and zero on $[\ell, L]$, and is bounded between zero and one. By uniqueness for the Dirichlet problem, it equals u_δ in (3.7). Equivalently, it is the harmonic measure of Γ_δ in $\mathbb{D}_\delta \setminus [\ell, L]$; see Ransford (1995, Sections 4.1 and 4.3).

At $z = 0$, $0 < T_\delta(0) < k$, so (G.11) gives

$$U_\delta(0) = \mathbf{F} \left(\arcsin \frac{R_\delta - L}{k(R_\delta + L)} \mid k^2 \right),$$

which is real.

Derivation of the pointwise real formula. Fix $z \in \mathbb{D}_\delta \setminus [\ell, L]$ and choose a determination $U_\delta(z) = x + iy$ with $0 < x < \mathbf{K}(k^2)$, $|y| < \mathbf{K}(1 - k^2)$. Put $r = \operatorname{sn}^2(x \mid k^2)$, $s = \operatorname{sn}^2(y \mid 1 - k^2)$. The addition formulas (G.8) and the imaginary transformation (G.9), applied to (G.10), give

$$\begin{aligned} |T_\delta(z)|^2 &= \frac{k^2(r + (1 - r)s)}{1 - (1 - k^2r)s}, & |k^2 - T_\delta(z)|^2 &= \frac{k^2(1 - r + (1 - k^2)rs)}{1 - (1 - k^2r)s}, \\ |1 - T_\delta(z)|^2 &= \frac{1 - k^2r - (1 - k^2)s}{1 - (1 - k^2r)s}. \end{aligned}$$

In particular, we have

$$\frac{1 + |T_\delta(z)|^2 - |1 - T_\delta(z)|^2}{k^2 + |T_\delta(z)|^2 + |k^2 - T_\delta(z)|^2} = r. \quad (\text{G.13})$$

Since $\operatorname{Re} T_\delta(z) > 0$ and

$$(1 + |T_\delta(z)|^2)^2 - |1 - T_\delta(z)|^2 = 4(\operatorname{Re} T_\delta(z))^2,$$

rationalizing the numerator in (G.13) shows that the square of the right-hand side of (3.9) equals r . Hence $b_\delta(z) = \operatorname{sn}(x \mid k^2)$ and $0 < b_\delta(z) < 1$. Real inversion in the first identity of (G.8) yields

$$\operatorname{Re} U_\delta(z) = \mathbf{F}(\arcsin b_\delta(z) \mid k^2). \quad (\text{G.14})$$

Substitution of (G.14) into (G.12) proves (3.10). As z approaches the slit, $x \rightarrow \mathbf{K}(k^2)$ and hence $b_\delta(z) \rightarrow 1$; as z approaches Γ_δ , $x \rightarrow 0$ and hence $b_\delta(z) \rightarrow 0$. This proves the stated continuous boundary values, including the two endpoints by continuity of the inverse map.

The complex function q_δ and its period. Start with the real value of U_δ at zero in (G.11) and continue it locally along paths in the slit disk. On every such local branch, (G.12) gives the continued harmonic conjugate with $\tilde{u}_\delta(0) = 0$ as $u_\delta(z) + i\tilde{u}_\delta(z) = 1 - \frac{U_\delta(z)}{\mathbf{K}(k^2)}$. Therefore (3.11) is equivalently

$$q_\delta(z) = \frac{\pi\{\mathbf{K}(k^2) - U_\delta(z)\}}{\mathbf{K}(1 - k^2)}. \quad (\text{G.15})$$

Changing U_δ by $2ji\mathbf{K}(1 - k^2)$ changes q_δ by $-2j\pi i$. Hence the local determinations of q_δ agree modulo $2\pi i$, and its real part is the one stated in the lemma. Formula (G.1) follows from the preceding value of $U_\delta(0)$.

On the slit, $\operatorname{Re} U_\delta = \mathbf{K}(k^2)$, so the two boundary values of q_δ are purely imaginary. Conjugation symmetry of (G.10) gives

$$q_{\delta,-}(x) = -q_{\delta,+}(x) \pmod{2\pi i}, \quad \ell < x < L.$$

The level curve and the exponent. The equation $u_\delta(z) = \delta$ is equivalent to

$$\operatorname{Re} U_\delta(z) = (1 - \delta)\mathbf{K}(k^2).$$

Using (G.10) and solving the definition of T_δ for z , a full parametrization is

$$z_\delta(\theta) = L - R_\delta \frac{1 - k \operatorname{sn}((1 - \delta)\mathbf{K}(k^2) - i\theta\mathbf{K}(1 - k^2)/\pi \mid k^2)}{1 + k \operatorname{sn}((1 - \delta)\mathbf{K}(k^2) - i\theta\mathbf{K}(1 - k^2)/\pi \mid k^2)}, \quad 0 \leq \theta < 2\pi. \quad (\text{G.16})$$

The map $z \mapsto U_\delta(z)$ is conformal after identifying values that differ by $2i\mathbf{K}(1 - k^2)$; hence (G.16) is an analytic Jordan curve. The line $\operatorname{Re} U_\delta = \mathbf{K}(k^2)$ maps to the slit. The inverse map in (G.16)

is analytic with bounded derivative when $\operatorname{Re} U$ lies in a sufficiently small fixed neighborhood of $\mathbf{K}(k^2)$ and $\operatorname{Im} U$ is taken modulo $2\mathbf{K}(1 - k^2)$, uniformly under (2.2). Comparing the two vertical lines therefore gives the Hausdorff distance

$$\operatorname{dist}_H(\Lambda_\delta, [\ell, L]) \leq C\delta.$$

Moreover, zero has real coordinate $U_\delta(0)$, and $U_\delta(0) < (1 - \delta)\mathbf{K}(k^2)$, exactly when $\delta < u_\delta(0)$. The latter inequality follows below for all sufficiently small δ . Thus the bounded component of the complement of Λ_δ contains the slit but not zero.

Set $\tau = \gamma(1 + \delta)^2$. Since $R_\delta = L\sqrt{\tau}$ and $\ell = L/\kappa$,

$$k = \frac{\sqrt{\tau} - 1 + \kappa^{-1}}{\sqrt{\tau} + 1 - \kappa^{-1}}, \quad \frac{R_\delta - L}{k(R_\delta + L)} = \frac{(\sqrt{\tau} - 1)/(\sqrt{\tau} + 1)}{k}.$$

Comparing (3.10) with the definition of α in Section 2 gives

$$u_\delta(0) = \alpha(\kappa, \tau) = \alpha(\kappa, \gamma(1 + \delta)^2). \quad (\text{G.17})$$

On $\{z : \delta < u_\delta(z) < 1\}$, the function $(u_\delta - \delta)/(1 - \delta)$ is harmonic, equals zero on Λ_δ , and equals one on Γ_δ . By the characterization of harmonic measure by the Dirichlet problem (Ransford, 1995, Section 4.3), this function is the harmonic measure of Γ_δ in that region. Evaluating at zero and using (G.17) proves the exact identity in (3.8).

Under (2.2), γ is eventually bounded above and bounded away from one. For fixed $\kappa > 1$, the arguments and parameters of the real elliptic integrals in (G.17) then remain in compact subsets of $(0, \pi/2)$ and $(0, 1)$. Differentiation under the defining integral for $\mathbf{F}$ and continuity of $\mathbf{K}$ therefore give

$$u_\delta(0) = \alpha(\kappa, \gamma) + O(\delta),$$

uniformly for all sufficiently large n . Substitution into the definition of α_δ proves the last equality in (3.8). Compactness also gives the three two-sided bounds stated in the lemma and, in particular, $\delta < u_\delta(0)$, for all sufficiently small δ . Letting $\delta \downarrow 0$ in (G.17) gives $u_0(0) = \alpha(\kappa, \gamma)$. Thus (3.10) gives the harmonic measure at zero asserted in the main text.

The contour formula. For $\mu(t) = L + R_\delta e^{it}$, we have $T_\delta(\mu(t)) = i \cot(t/2)$. For $t \notin 2\pi\mathbb{Z}$, let

$$y(t) = \mathbf{F} \left(\arctan \frac{\cot(t/2)}{k} \mid 1 - k^2 \right),$$

as in Section 3.2. Since the angle in the last display lies in $(-\pi/2, \pi/2)$, the inverse identity and (G.9) give $k \operatorname{sn}(iy(t) \mid k^2) = i \cot(t/2)$. Thus one determination on the outer circle is $U_\delta(\mu(t)) = iy(t)$, and (G.15) gives

$$q_\delta(\mu(t)) = \frac{\pi\{\mathbf{K}(k^2) - iy(t)\}}{\mathbf{K}(1 - k^2)} \pmod{2\pi i}.$$

As t increases counterclockwise from 0 to 2π , $y(t)$ decreases from $\mathbf{K}(1 - k^2)$ to $-\mathbf{K}(1 - k^2)$. Hence q_δ increases by $2\pi i$, fixing the sign of the period asserted above. Finally, we have

$$q_\delta(0) = \frac{\pi\mathbf{K}(k^2)}{\mathbf{K}(1 - k^2)} u_\delta(0),$$

so substitution into (3.15) gives exactly (3.22). At $t \in 2\pi\mathbb{Z}$, the two one-sided values of q_δ differ by $2\pi i$; because K is an integer, they give the same value of g_K .

The argument used only the inequalities $R_\delta > L > \ell > 0$ until the model-specific calculation (G.17). Therefore the coordinate, period, and formula (G.1) remain valid when R_δ, L, ℓ are replaced by any positive numbers R, L, ℓ with $R > L > \ell$. This is the form used for the analogous potential in Appendix F. $\square$

H Supporting analytic and probabilistic estimates

H.1 Known covariance, approximation, and smooth calculus

For the known-covariance estimator, put $w = \Sigma^{-1}v$. Then $\mathbb{E}[w^\top X_i Y_i] = \beta_v$ and $\|w\|_2 \leq \ell^{-1}$. Independence, the fourth-moment bound for standardized linear combinations of independent entries, and bounded $\|\Sigma\|_{\text{op}}$ give $\mathbb{E}(w^\top X_i)^2 (\beta^\top X_i)^2 \leq C$. The error term contributes $\sigma^2 w^\top \Sigma w \leq C$. Thus averaging gives mean squared error at most C/n . A pair $\beta = \pm cv/\sqrt{n}$ at a fixed Gaussian covariance and noise variance gives its matching root- n testing scale by the elementary likelihood calculation. This comparison does not assume an estimable full precision matrix when that matrix is unknown.

Lemma H.1 (Analytic polynomial approximation; [Trefethen, 2013](#), Theorem 8.2). Let $\rho_0 > 1$ and let E_{ρ_0} be the filled ellipse with boundary $\{(w + w^{-1})/2 : |w| = \rho_0\}$. If f is analytic on a neighborhood of E_{ρ_0} and $\sup_{E_{\rho_0}} |f| \leq M_0$, its degree- M truncated Chebyshev expansion p_M satisfies

$$\sup_{x \in [-1, 1]} |f(x) - p_M(x)| \leq \frac{2M_0 \rho_0^{-M}}{\rho_0 - 1}.$$

Lemma H.2 (Smooth functional calculus from a resolvent comparison). Suppose the difference of two expected self-adjoint resolvents has operator norm at most $C\epsilon |\text{Im } z|^{-s}$ for a fixed integer $s \geq 1$. Then every compactly supported smooth real function f satisfies

$$\|\mathbb{E}f(A) - \mathbb{E}f(B)\|_{\text{op}} \leq C_s \epsilon \|f\|_{W^{s+1, 1}}.$$

Remark H.3. The matrix-valued resolvent estimate of [Bandeira et al. \(2023\)](#), Theorem 2.8) gives this bound with $s = 5$. The scalar-trace version is [Bandeira et al. \(2023\)](#), Lemma 5.11). We use its almost-analytic integral argument for the operator-norm comparison. The same integral representation is recorded in [Yang \(2024\)](#), Lemma 3.8).

Proof. Choose a smooth compactly supported $\chi(y)$ equal to one on $|y| \leq 1$ and form $\tilde{f}(x + iy) = \chi(y) \sum_{j=0}^s f^{(j)}(x) (iy)^j / j!$. Differentiation gives $|\partial_{\bar{z}} \tilde{f}(x + iy)| \leq C_s |y|^s |f^{(s+1)}(x)|$ near $y = 0$. On the support of χ' , this derivative is bounded by $C_s \sum_{j=0}^s |f^{(j)}(x)|$. Cauchy's integral formula and Green's theorem, first for a scalar real spectral value and then integrated against the spectral measure, give

$$f(A) = -\frac{1}{\pi} \int_{\mathbb{C}} \partial_{\bar{z}} \tilde{f}(z) (z - A)^{-1} dA(z).$$

The scalar identity follows by deleting a small disk about the spectral value and letting its radius tend to zero. Subtract the two formulas and apply the assumed resolvent bound. The factor $|y|^s$ cancels its singularity near zero; the region where $\chi' \neq 0$ is separated from zero. Integrating in x and y proves the bound. $\square$

Lemma H.4 (Conditioning a mixing distribution). Let π be a probability measure on a measurable parameter space $(\Theta, \mathcal{T})$, and let M be a Markov kernel from $(\Theta, \mathcal{T})$ to a measurable observation space $(\mathbf{X}, \mathcal{A})$. For $\mathcal{C} \in \mathcal{T}$ with $\pi(\mathcal{C}) > 0$, define the mixture laws by

$$(\pi M)(A) = \int_{\Theta} M(\theta, A) \pi(d\theta), \quad (\pi(\cdot \mid \mathcal{C})M)(A) = \frac{1}{\pi(\mathcal{C})} \int_{\mathcal{C}} M(\theta, A) \pi(d\theta), \quad A \in \mathcal{A}.$$

We have $\text{TV}(\pi(\cdot \mid \mathcal{C})M, \pi M) \leq \pi(\mathcal{C}^c)$.

Proof. If $\pi(\mathcal{C}) = 1$, the two mixture laws coincide. Otherwise, both conditional mixing distributions are well defined, and splitting the integral over $\mathcal{C}$ and $\mathcal{C}^c$ gives

$$\pi M = \pi(\mathcal{C}) \pi(\cdot | \mathcal{C})M + \pi(\mathcal{C}^c) \pi(\cdot | \mathcal{C}^c)M.$$

Subtracting this identity from $\pi(\cdot | \mathcal{C})M$ and taking the supremum over measurable observation events yields

$$\text{TV}(\pi(\cdot | \mathcal{C})M, \pi M) = \pi(\mathcal{C}^c) \text{TV}(\pi(\cdot | \mathcal{C})M, \pi(\cdot | \mathcal{C}^c)M) \leq \pi(\mathcal{C}^c),$$

because the total variation distance between two probability measures is at most one. $\square$

Lemma H.5 (Matrix-volume integral). For $t > 0$, integrate over real symmetric positive definite $s \times s$ matrices with Lebesgue measure $d\mathbf{Q} = \prod_{1 \leq i \leq j \leq s} d\mathbf{Q}_{ij}$. Then

$$\int_{\mathbf{Q} \succ 0} \exp(-t \text{tr}(\mathbf{Q} - I_s - \log \mathbf{Q})) d\mathbf{Q} = e^{ts} t^{-s(t+(s+1)/2)} \pi^{s(s-1)/4} \prod_{j=1}^s \Gamma\left(t + \frac{s+2-j}{2}\right).$$

For $t \asymp d$ and $s = O(\log d)$, the absolute value of the logarithm of this integral is at most $Cs^2 \log d$.

Proof. Write $\mathbf{Q} = TT^\top$ with T lower triangular and positive diagonal. Differentiating its independent entries successively gives the Jacobian $2^s \prod_{j=1}^s T_{jj}^{s+1-j}$ for the stated measure. Each off-diagonal integral equals $\sqrt{\pi/t}$. Each diagonal integral, after the substitution $y = tT_{jj}^2$, equals

$$\frac{1}{2} t^{-(t+(s+2-j)/2)} \Gamma\left(t + \frac{s+2-j}{2}\right).$$

Multiplying these factors, the Jacobian constant, and e^{ts} proves the formula. In Stirling's formula the terms of orders $st \log t$ and st cancel the displayed power and exponential. Each of the s gamma arguments differs from t by $O(s)$, so all remaining terms are bounded by $Cs^2 \log d$. This proves the asserted estimate. $\square$